USING MACHINE LEARNING TO PROVIDE AN OBJECTIVE MEASURE OF SPEECH INTELLIGIBILITY IN DYSARTHRIC SPEAKERS

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Abstract

Current subjective means of clinical assessment are both time and labour intensive, often placing immense burden and strain on vulnerable persons. Developing an objective method of speech intelligibility in dysarthric persons would be an indispensable tool to clinical professionals monitoring speech symptoms in the long term. By evaluating frameworks proposed in previous works and continuing their research, this paper introduces novel features and subsystem developments to improve upon existing results. Spectral features, such as Spectral Tilt and Harmonic-to-Noise Ratio, are extracted from single-word utterances. The addition of these features, in concert with a multiple-classifier approach, provide a classification accuracy of 82% when assessing the severity of a speaker's dysarthria. An objective estimate of the speaker's intelligibility is then reliably calculated using a polynomial regression model with an RMSE error of just 6.11. This improvement over previously published works provides a bright outlook for the field going forward, with some potential future developments being developed and discussed in this paper.

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1 . Introduction

Dysarthria is a motor-speech disorder acquired as the result of neurological injury or changes to the brain in later life [1]. It is also a common symptom of Parkinson's Disease (PD), a type of neurodegenerative disease associated with dopamine deficiency, and Cerebral Palsy (CP), a group of lifelong neuromotor disorders that affect movement and coordination [2]. Dysarthric speech can present differently, with symptoms dependent on the etiology, but it is thought to be the most common speech disorder in the world [3]. In all cases, it can present a barrier to communication and have a hugely adverse effect on the speakers' lives. The specific features of dysarthric speech are dependent on the exact cause; speakers with PD and CP could both present with dysarthric speech and sound completely different to one another. For example, the Parkinsonian dysarthric voice is often characterised as being hoarse, excessively nasal and low volume, whereas, Cerebral Palsic speakers are said to have strained phonation, imprecise placement of articulators and incomplete consonant closure [4].

Currently, speech pathologists rely on subjective speech assessments to characterize the severity of speech disorders, to monitor and document the progression of PD or other dysarthria causing diseases, and to plan treatments [5][6]. These subjective speech assessments are costly, time-intensive, and very susceptible to biases, such as familiarity with the patient and the speech pathologist providing the assessment. Providing a machine learning alternative could provide the opportunity for more regular speech intelligibility exams which are objective, consistent, instantaneous, and could be conducted in the patients home through their smartphone [6], [7].

This report describes the design and implementation of a machine learning system that aims to provide objective estimates of speech intelligibility for people with speech dysarthria, primarily Cerebral Palsy. The initial intention of the project was for the system to be used for people diagnosed with Parkinson's disease and it will be explained why this was the initial intention, as well as why the aims of the project were shifted.

1.1 . Project Aims

The aims of the project work are as follows:

1. To create and train a machine learning model which provides an objective estimate of speech intelligibility for dysarthric persons where speech recordings are taken from a previously collected dataset.

2. To explore techniques from literature, as well as novel ideas, to improve upon previous works in this area and create a competitive system. This includes the extraction of previously unused audio features as well as modifications to previously used algorithms for speech intelligibility estimation.

3. To analyse the potential for clinical use of the developed system or similar systems in the future.

1.2 . Structure of the Report

Chapter 2 contains background information that informs the direction of the project. Part of this is from an initial review of the current literature, though this has been expanded significantly throughout the duration of the project, especially as the course of the project has changed throughout its duration. It begins with an overview of dysarthric speech and the clinical needs which inspired this project, before moving into the current state of machine learning for audio systems and how these are applied to dysarthria.

Chapter 3 describes the dysarthric speech datasets utilised within this project, as well as justifications for their use and disadvantages for each.

Chapter 4 documents the process of extracting features from the speech recordings and provides the correlation scores for each feature with the provided subjective speech intelligibility rating for each subject in the dataset. The reasoning for the choice to use each feature is provided, as well as reflection on the effectiveness of each chosen feature.

Chapter 5 details the creation of the binary classification system used to initially label a speech recording as *low-mid* or *mid-high* intelligibility. The method for implementing, training and combining each classifier into a novel ensemble algorithm is described.

Chapter 6 describes the process of creating and evaluating the system for objectively estimating speech intelligibility for each subject in the data set. Evaluation of the system is compared against previous works with the same goal and against the potential for clinical use. Also included is a discussion of some evaluation criteria missing from previous works and the impact of these metrics on a system's clinical viability.

Chapter 7 talks about the process of assembling the subsystems to carry out further tests and provide a final evaluation. Experiments are carried out to provide a fuller analysis of the system as a whole and compare the performance to those of previous works.

Chapter 8 evaluates the system as a whole and compares it to previous research, as well as the initial goals of the project. Reflection on the current state of the system when compared to the initial goals shows how the goals of the system have been achieved or shifted throughout development.

Chapter 9 briefly discusses and reflects on the management of the project. This paper describes the major research project that has been undertaken and the substantial work of code which has been created over several months. Time management over this period, including project schedules, are included here.

Chapter 10 is the final chapter and discusses potential for future research beyond the project described here. Several pilot experiments are undertaken to evaluate the current performance of the system in new circumstances and assess the need for further study to prepare the system for clinical application.

2 . Background

To develop a system capable of objectively rating dysarthric speech, it is crucial to understand the mechanisms through which dysarthrias affect speech and how affected speech presents quantifiably in recorded audio. The production of speech by the human body is a complex process, involving many different muscles and chambers within the body whose dimensions affect speech in different ways. Without proper understanding, it is easy to take standard acoustic measurements from a recording which do not relate to dysarthria, leading to inconsistent and false results. Furthermore, understanding of the theory behind machine learning algorithms and other data analysis techniques will be researched to ensure that good data is followed by good analysis, providing the basis for confidently presenting a system for real-world use.

2.1 . Parkinson's disease

Parkinson's disease is a type of neurodegenerative disease associated with dopamine deficiency [8]. Dopamine is a hormone that acts as a transmitter between nerve cells in the brain, and people with decreased levels of dopamine suffer deterioration in various activities controlled by the brain. Decreased levels of dopamine in PD patients is caused by damage to the substantia nigra, a part of the brain which secretes dopamine [9], [10]. This can be seen in Figure 1, where a PD patient is shown to have atypical pigmentation loss in the substantia nigra [11].

The symptoms with which this presents physically in patients is first usually muscular tremors, hypokinesia (lack of movement) and bradykinesia (slowness of movement). As the muscular symptoms occur throughout the body, the muscles involved in producing speech are also affected by the disease. Patients with PD can often have difficulty articulating phonemes and speak with weaker voices with more breathy qualities [12].

While there is currently no cure for PD, there are treatments available that can extend the life of Parkinson's patients, as well as improving their quality of life, making early detection crucial. One such available treatment to reduce the motor symptoms of PD is Deep Brain Stimulation.

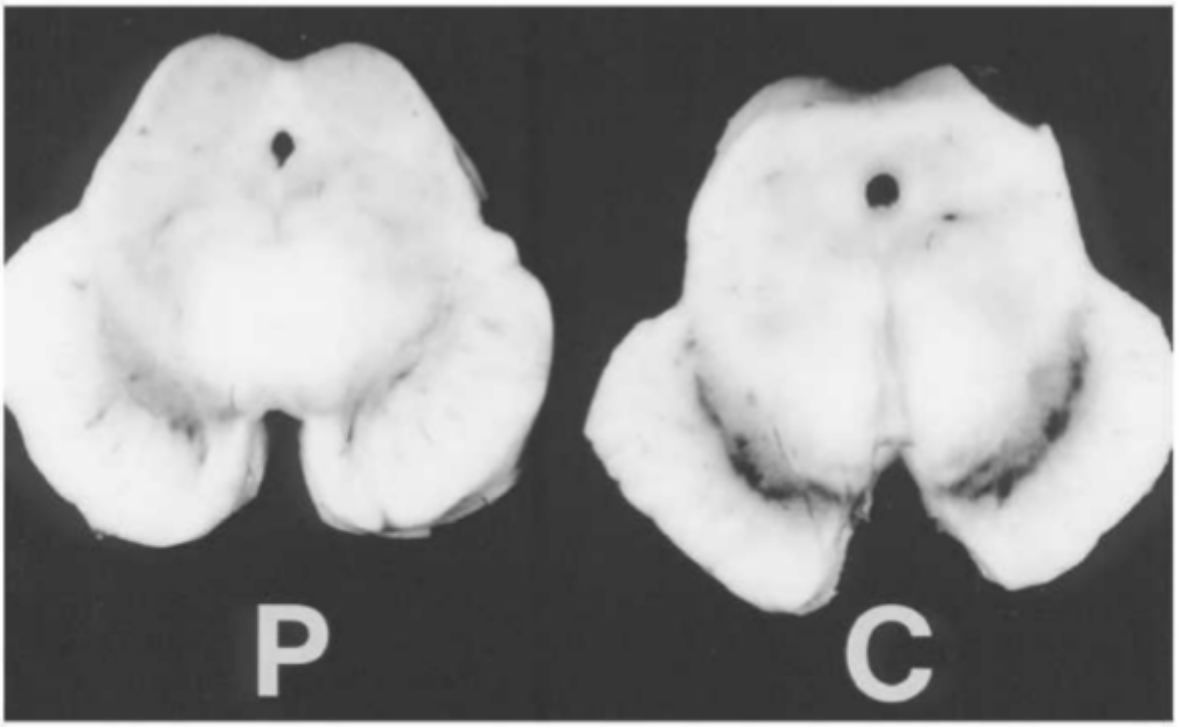

Figure 1. Horizontal sections of midbrain from a patient with PD (P) and a normal control (C), showing loss of pigmentation in the substantia nigra in PD [11].

2.1.1 . Deep Brain Stimulation

Deep Brain stimulation (DBS) involves placing electrodes into the brain, which are attached to a pulse generator, to reduce motor fluctuations and refractory tremors [12]. Patients who undergo surgery to receive a DBS implant often experience a significant reduction in these motor symptoms [13].

While DBS is proven to help manage many of the physical symptoms of PD, there have been concerns noted over the effect of DBS on patients' speech. There is some debate about whether DBS actively aggravates speech symptoms or has no effect on speech at all, allowing speech symptoms to worsen with the natural progression of the disease [14], [15]. In either case, people who undergo DBS frequently still report issues with their speech. Furthermore, some studies have shown that DBS can have other neurocognitive effects including increased apathy scores and depression [16].

2.2 . Cerebral Palsy

Cerebral Palsy (CP) is an umbrella term for a group of permanent but non-progressive motor disorders, developed as a result of brain lesions or abnormalities in early development [17]. It is classified into three main groups: spastic, ataxic and dyskinetic. These types present with different symptoms and people with CP can also be classified as having a mix of these types. The severity of the disorder varies massively from case to case and symptoms can go unnoticed until early childhood, around age 2 - 3 [18]. Most of the studies in western countries showed a prevalence of CP at around 2–2.5 per 1000 live births, making it the most common cause of motor disorder in early childhood [19], [20]. Many people with CP also present with other associated impairments such as intellectual impairment, spastic dysarthria, sensory impairment or epilepsy [21]. The initial neurological cause of the disorder does not worsen in CP cases, though the symptoms of the disorder can place excess strain on the body and cause joint pain and other issues in later life. Mental health can also be affected in children and adults as they struggle with communication [22].

In children, CP is the most prevalent severe disability after severe intellectual impairment. It is twice as common as Down syndrome and it accounts for 60% of severe motor disabilities [23]. Most children with CP require some specific help during their schooling period, with limitations on their independence often following them into adulthood.

2.2.1 . Speech Symptoms of Cerebral Palsy

While not all CP cases demonstrate badly diminished speech, speech dysarthria is one of the most common symptoms, with up to 90% of CP cases showing dysarthric speech [24]. Dysarthria varies by CP type, just as it varies with different causal motor disorders. For example, Individuals with Parkinson's disease (PD) exhibit hypokinetic dysarthria, which is a motor speech disorder caused by damage to the basal ganglia control circuit [25]. The basal ganglia also happens to be the macrostructure containing the substantia nigra, the part of the brain most associated with Parkinson's Disease. In contrast, the most common form of dysarthria diagnosis in persons with CP is spastic dysarthria, associated with spastic-type CP, the most common form of CP [26][27]. Figure 2 outlines the main cerebral palsy types and their associated brain regions [27]. In all cases, dysarthria in CP is caused by impaired neuromuscular control over the speech mechanism and the severity of the dysarthria increases with the severity of motor CP symptoms [28], [29].

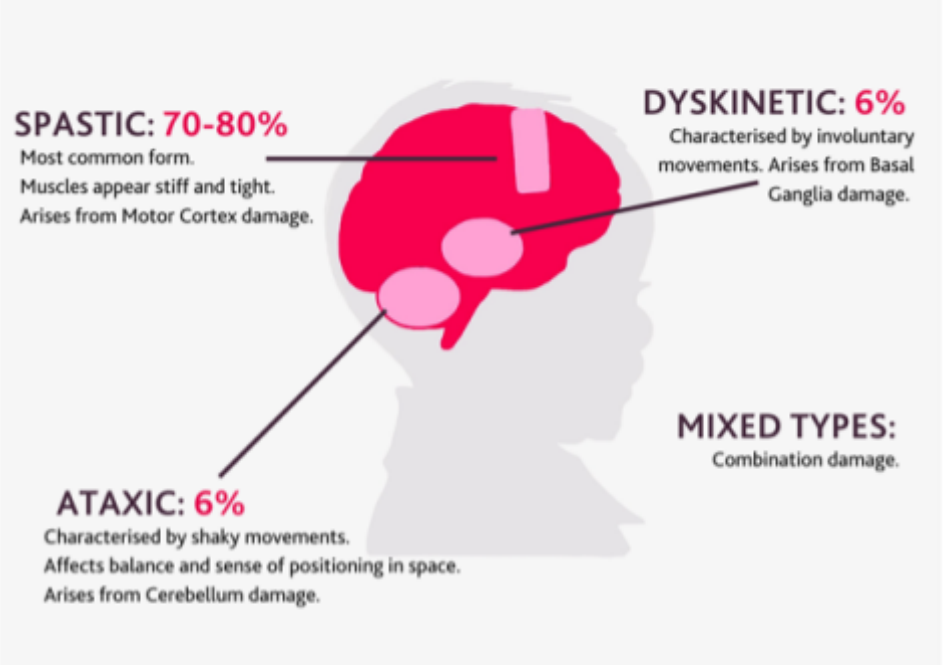

Figure 2. The types of Cerebral Palsy and the brain regions associated with each diagnosis. [27]

The effect of dysarthria in children with CP reaches further than just difficulty articulating phonemes. Children with CP show reduced participation in conversation with adults, rarely starting conversations and often taking a purely respondent role [30]. The effect of diminished communication in young children with CP has been the subject of many studies investigating the effect on intelligence and mental health in later life [31].

2.3 . Clinical Need for an Automatic Speech Quality Estimation System

Currently, diagnosing and observing the progression of disordered speech is a manual and time-intensive process, and it has been pointed out that there is a need to move to more objective data as a basis for making clinical decisions [32]. Relying solely on in-person subjective analysis leaves much more room for external factors to influence the data, such as the clinician's prior knowledge of the patient introducing bias, or changes in the patient's symptoms as a result of travel to the clinical office. Furthermore, required physical presence at a hospital for these examinations can be logistically and cost-prohibitive for both the clinician and patient.

By taking objective measurements or speech recordings, remote analysis without the need for a speech pathologist on-site is potentially possible. Much more data can be collected and analysed, with the potential for extra measurements to be retroactively taken if speech recordings are stored. By applying phoneme analysis on speech recordings in dysarthric speakers, exact phoneme difficulties can be identified, meaning a more specific speech therapy can be prescribed.

2.3.1 . Criticisms and Viability Concerns

It has been observed that some audio features and ML systems are not robust to noise and classification can be dependent on the audio recording environment [33]. The study undertaken in [7] involved the creation of a new data set containing audio recordings using a professional-grade cardioid microphone and a microphone built into a smartphone. Doing this allowed them to not only use the data to assess the accuracy of their classification method but also assess the viability of smartphones, which are widely available, as a recording device for audio to be classified. It was shown that there was no significant difference between the two audio recording methods in terms of classification, opening the doors for smartphones to be potentially used in a clinical setting and reducing the barrier to entry for an ML-based classification system to be deployed.

2.4 . Machine Learning Classification and Regression Algorithms

Machine learning is an extremely popular and fast-growing area of data analysis. The basis for machine learning is the use of statistical methods and algorithms to identify patterns in training data, and the use of these patterns to make predictions about unseen data. In the simplest systems, statistical techniques can group data based on similarity. In larger systems, patterns can be abstracted further from the original data, leading to almost human-like recognition and beyond.

There is debate in the research community about the use of the term ‘machine learning', and whether several techniques used here, such as support vector machines, could perhaps be more accurately described as statistical algorithms. For the purpose of this paper, however, all classification and regression algorithms used will be described as machine learning techniques.

2.4.1 . Support-Vector Machine

One of the most common types of classifiers used in previous literature is the Support-Vector Machine (SVM). This classifier can be seen in previous implementations of speech intelligibility estimation systems as well as other works analysing dysarthric speech data [34]–[37]. The discriminatory mechanism behind an SVM involves plotting the labelled data points in a high-dimensional feature space, then algorithmically determining a plane or hyper-plane which successfully encompasses or separates each labelled group. An example of this is shown in Figure 3 [38]. To determine the optimality of a hyperplane separating two or more groups, the margin distance between the proposed hyperplane and the closest members of each group is maximized, theoretically providing the most reliable separation between groups.

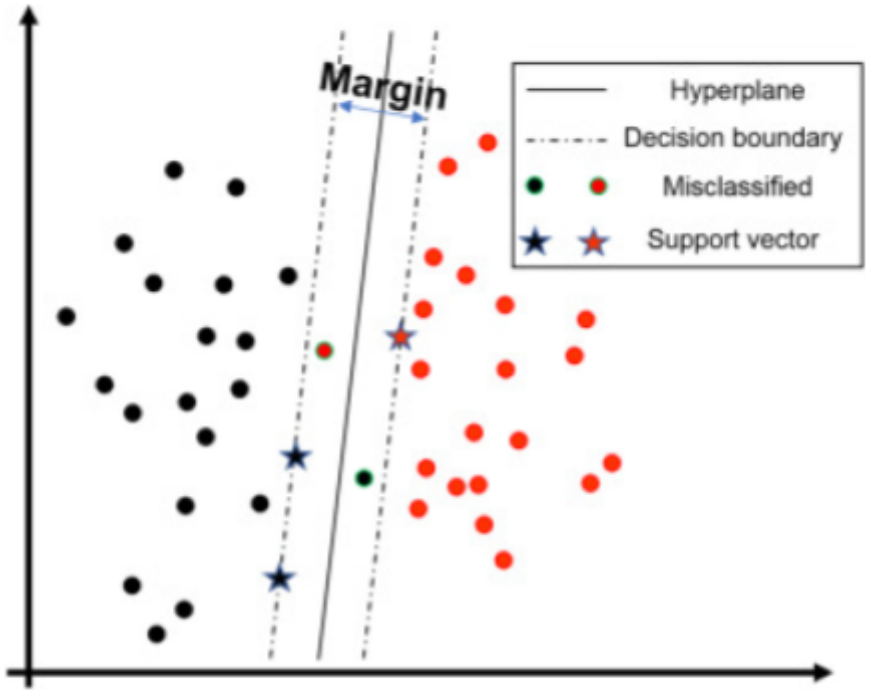

Figure 3. A diagram of a binary 2-feature SVM showing the hyperplane and decision boundary [38]

This simple idea can be extended to n-dimensional feature space, as well as non-linear feature space, with Radial Basis Functions (RBFs) being very popular.

2.4.2 . Multi-Layer Perceptrons

A Multi-Layer Perceptron (MLP) classifier is a type of feed-forward artificial neural network with at least three layers of perceptrons. A single perceptron is a type of binary classifier that utilises one of many potential activation functions, such as basic thresholds or sigmoid curves, to classify an input as belonging to a class or not. These perceptrons are interconnected and arranged in at least three layers: a hidden layer, an information layer, and a yield layer. The network is then provided with data, and a learning algorithm is applied to the system as a whole to optimise performance by algorithmically adjusting the weights of individual perceptrons.

Networks like these are capable of recognizing information not readily available from the input data, leading to unexpectedly complex behaviours from a network of many individually simple nodes. They have been largely eclipsed by neural networks in modern data science, but often appear when used as a smaller network with only one hidden layer [35]. An example of an MLP with a single hidden layer is shown below in Figure 4 [39].

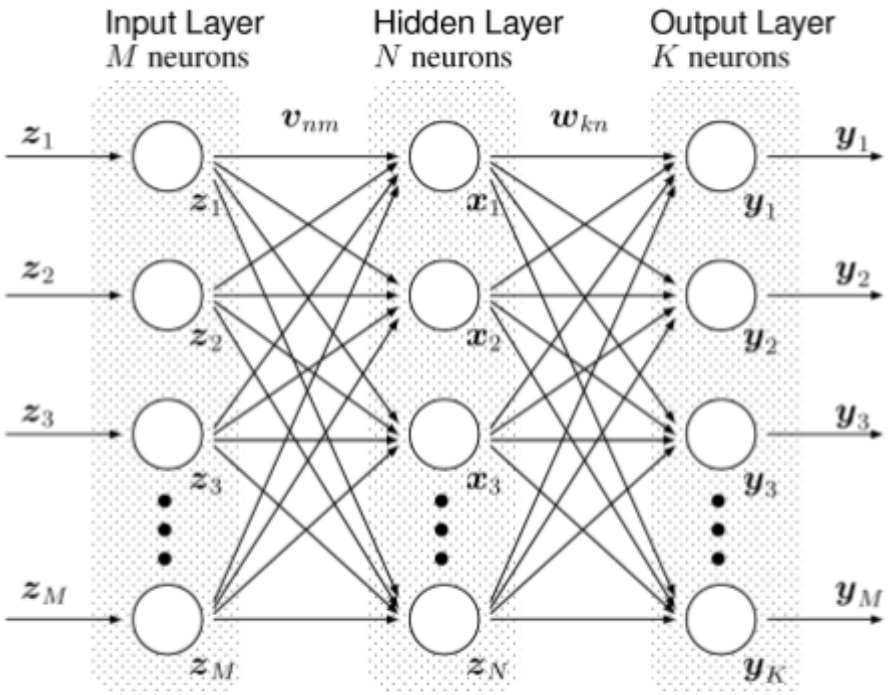

Figure 4. A simple MLP with one input layer, one hidden layer and one output layer [39].

2.4.3 . Decision Tree

A Decision Tree (DT) is one of the simplest structures for a machine learning system. A DT is a branching network of nodes at which simple evaluations are made on the data, which are connected through edges representing the decision made at each node. Structurally, this might resemble a large set of *if-else* statements. What elevates this to a machine learning algorithm is the process through which a tree is designed.

A root node is chosen by iterating through potential decisions to arrive at a final decision that provides the greatest individual information gain. The tested decisions cover all provided features and will test each feature in several different ways to retrieve the informational importance of the feature at that point in the tree. Once a root node is chosen, the process is applied recursively to every branch until the stopping criteria are met. Usually, the criteria for stopping the recursion are that all branches have connected to a single output node or a maximum recursion depth has been reached. A simple decision tree is shown in figure 5 [40].

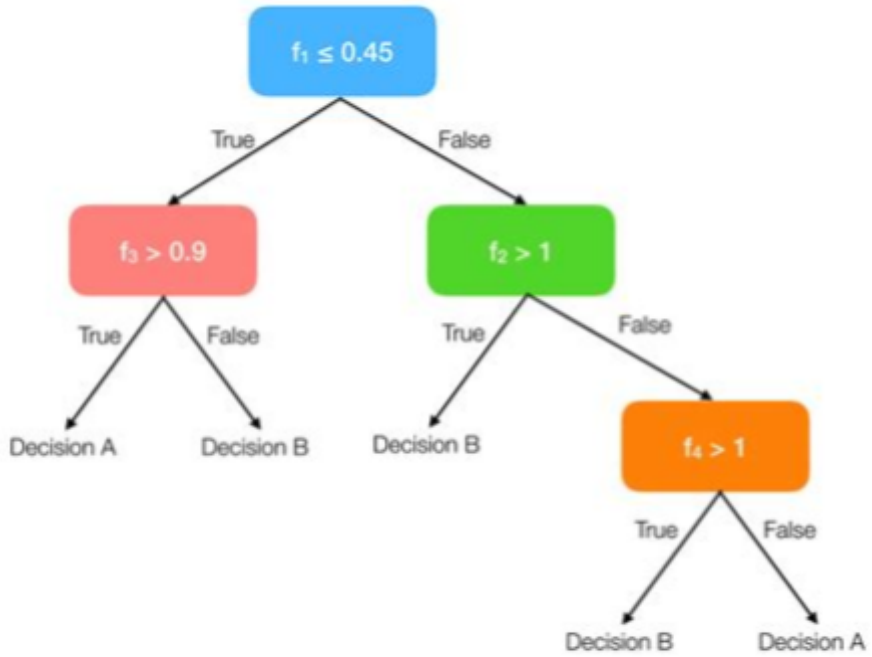

Figure 5. A simple binary decision tree [40]

Decision trees are frequently used on datasheets where a small number of features are used and/or transparency in the model is required. Due to the nature of DTs, the importance of each feature and its effect on the output can be easily visualized and understood, unlike some more 'black-box' style models like neural networks, for example.

2.4.4 . Random Forest

Random Forest is an example of an ensemble algorithm, where several algorithms or several instances of the same algorithm are implemented, with their outputs being combined to produce a final result. In this case, random forest implements several Decision Trees, with each tree being generated and trained using a random subset of the training data. Each tree is trained separately but all trees are deployed in parallel. Upon deployment, every tree in the forest is provided with the same test data and each tree comes to its own determination as to the classification label of the test data. Votes from each tree are then aggregated to produce a final result. Typically a simple majority vote is used, but it is possible to apply weightings to each tree's vote depending on the error rate of each tree. Figure 6 shows the structure of a simple random forest composed of n decision trees and a majority vote [41].

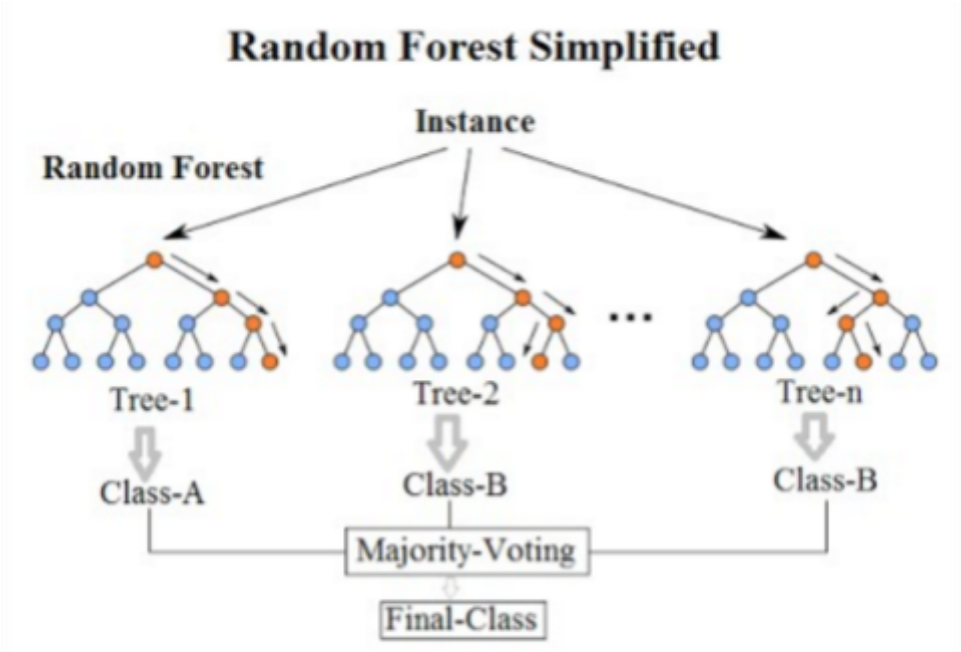

Figure 6. An n-tree random forest with a majority vote [41].

Using random forest as an alternative to simple decision trees increases the computational complexity of the training and testing proportionally to the number of trees. The results are also much harder to interpret than a single tree, leading some to describe Random Forest as a more black-box approach.

However, this can be a useful trade-off as the classification result is effectively averaged over a larger set, reducing the effect of minor noise in individual trees. Furthermore, since any one tree only sees a subset of the training data, there is a much smaller risk of overfitting, which can be otherwise challenging to recognise and mitigate in other ML systems.

2.4.5 . Adaptive Boost Classification

Adaptive Boost is another type of ensemble algorithm, combining many identical weak classifiers to produce a more accurate overall classification. Unlike Random Forest, the training data subsets assigned to each weak classifier are not entirely random.

The subsets used to train each weak classifier are not all created at once. First, one truly random subset is derived from the complete training dataset. After training the first weak classifier, each item in the subset is assigned a weight, with misclassified items being weighted higher than correctly classified items. Each item is then added back to the complete training set and a new subset is selected to train the next weak classifier. However, it is now more likely for previously misclassified items to appear in subsequent subsets. This process is then repeated for each weak classifier. This algorithm allows the system to focus its attention on more difficult cases and specifically target areas for improvement.

Further to this, each classifier is also assigned a weight after it’s trained. Classifiers with 50% accuracy are assigned a weight of 0, higher accuracy classifiers are assigned a positive weight, and lower accuracy classifiers are assigned a negative weight. More highly weighted classifiers then affect the final output much more, leading to a more accurate overall system. Figure 7 outlines the algorithm, showing how initial weak classifiers weight the original data to create new weighted pools for the next semi-random selection [42].

Figure 7. An overview of the Ada Boost algorithm, using colour to demonstrate weighting adjustments [42].

2.4.6 . Gaussian Naive Bayes

Naive Bayes classification is a method of probabilistic classification in which the output is given by directly calculating the probability of a data point, with a set of features, belonging to a class. The class with the highest probability of the datapoint belonging to it is selected as the final output.

Truly calculating these probabilities would require a large amount of data and prior knowledge, hence a naive assumption is made that the initial probability for each class is equal to the class' distribution within the training data and all features are stochastically independent, given the class. Gaussian distribution is used to model features that aren't simply binary or integers.

2.4.7 . Gaussian Process

Gaussian Process classifiers (GPC) are an interesting implementation of a classification algorithm. The core of a binary GPC actually solves the problem by once again modelling the features as a high-dimensional multivariate Gaussian. However, rather than directly producing a class probability, it produces a decimal numerical output between 0 and 1 and thresholding the given value (usually at 0.5) to provide the final binary labels. This means that GPC is actually a Gaussian regression algorithm.

2.4.8 . Linear and Quadratic Discriminant Analysis

Quadratic Discriminant Analysis is an extension to Linear Discriminant Analysis, a probabilistic model in which it is assumed that the joint density of all features, conditional on the target's class, is a multivariate Gaussian.

In Linear Discriminant Analysis, it is assumed that the covariance matrix for each class is the same, resulting in a linear hyperplane separating each class similarly to linear SVM. Much like linear SVM, it’s clear that not all datasets are readily separable in a linear space. This is why Quadratic Discriminant Analysis is used to define a unique covariance matrix for each class, resulting in quadratic decision boundaries useful for separating nonlinear data with varied covariances.

2.4.9 . K Nearest Neighbours

The K Nearest Neighbors (KNN) algorithm can be used to classify data by analysing clustering patterns within a feature space, similarly to the SVM algorithm. The KNN algorithm determines the label of a new data point by plotting it in feature space and looking at the labels of the k most proximal training data points. The most popular label of the k nearest pre-existing points will be applied to the new data point. This distance-based classification approach is visualised in Figure 8 [43].

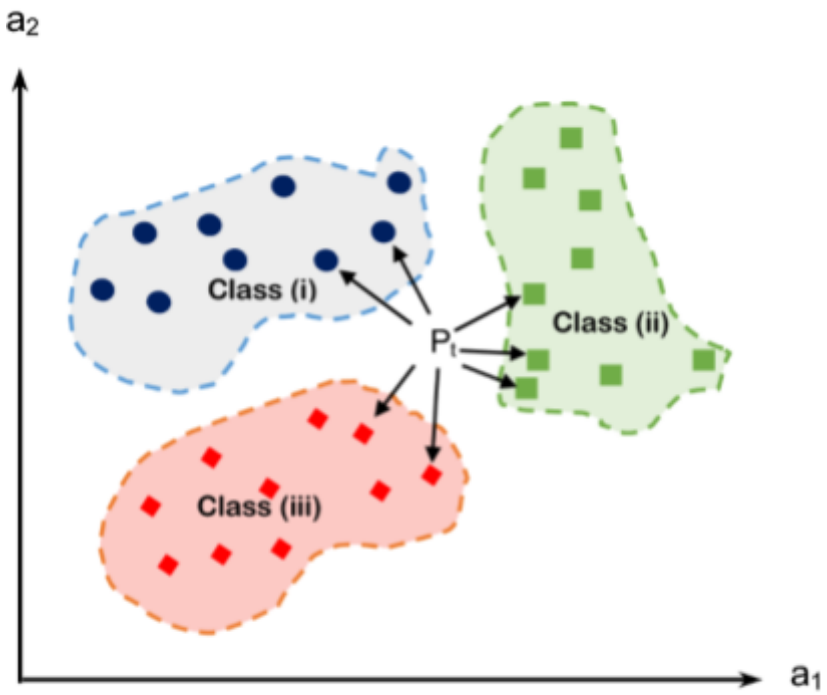

Figure 8. The placement of a new point (Pt) with its 7 nearest neighbours [43].

Two of the main practical differences between KNN and SVM are the ability to vary k to optimise performance and the consideration of proximity and centre of mass between groups, rather than defining binary boundaries based solely on the outermost members of a class. In SVM, the hyperplane is selected based on the margin between the closest points in two different groups, the distribution of points within the space is otherwise not considered, making the system potentially vulnerable to outliers and most similar to a KNN system where k=1. In KNN, the higher k is, the more proximal points will be considered, which would favour a denser but slightly further cluster over a sparse cluster with only a few points closer to the new data point.

A benefit to the KNN algorithm is the lack of training time. The only computationally taxing part of the algorithm is simply calculating the distance of the k nearest points to any new data point, a task that is only performed when classifying a new data point. The training process for the KNN algorithm is simply providing the data set, with extracted features.

2.4.10 . Polynomial Regression

In regression analysis, the goal is to model the relationship between some dependent value y in terms of a dependent variable (or vector of independent variables) x. Through various methods, formulae can be estimated to describe the relationship between variables and model future behaviours. In the case of general polynomial regression, the relationship is modelled using an nth degree polynomial as shown below in equation 1.

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots + \beta_n x^n + \varepsilon. \qquad (1)$$

Greater values of n provide greater flexibility in the model, allowing for the modelling of nonlinear data to a much higher degree. In practice, models with a value of n higher than 4 are rarely used, as this level of flexibility is prone to overfitting. It is also computationally very expensive at this point. As the number of weights (β_k) increases, computational complexity increases exponentially.

The choice of an appropriate degree to use for a dataset is made by training and validating the data with a range of values, to search for the value n which provides the lowest mean-squared error. The exact validation technique used is important, as hunting for a lower mean-squared error in this way can introduce overfitting.

2.5 . Overfitting

Overfitting is where the model performs extremely well with the training data but poorly with novel data because it has 'learned' too much about the training data and has developed too specific criteria for labelling data. This can lead to excellent performance with the training data and incredibly poor performance with unseen test data, as anything not perfectly matching the training data is potentially rejected. An illustration of this is shown in Figure 9 below [44]. The avoidance of overfitting is a wide area of study going far beyond the scope of this paper. However, some mitigation tactics have been used in this project.

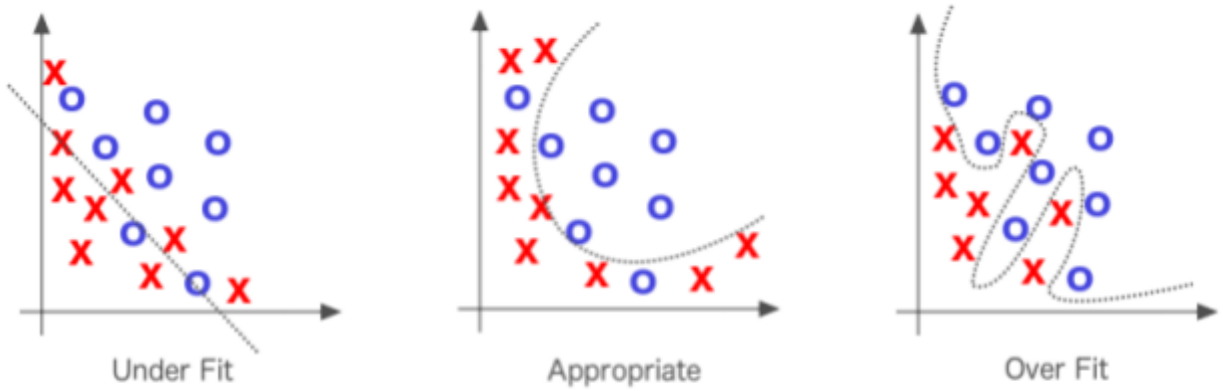

Figure 9. An illustration of a decision boundary for unde fit, over fit and appropriately fit classifier [44].

2.5.1 . Limiting Parameters and Ensuring Correlation

One of the simpler methods for avoiding overfitting is limiting the knowledge the algorithm has about the dataset and ensuring that provided data is correlated with the desired output. Correlation analysis like finding the Spearman's Rank Correlation Coefficient and Pearson's Rank Correlation Coefficient is commonly applied to features extracted from data before they are passed to a machine learning algorithm. Only highly correlated features are picked from a larger feature set, minimising unnecessary information and noise which the algorithm could erroneously attempt to use to learn about the data.

This is not always an easy process. Not all correlations can be quantified by techniques like these, and it may not be possible to extract several completely independent features from real data. After all, if it were always easy to use basic statistical correlation techniques to quantify relationships in data, it would be almost unnecessary to use machine learning at all.

2.5.2 . Validation Techniques

Cross-validation techniques are employed during the training phase to evaluate the development of ML systems. These techniques temporarily hide some portion of the training dataset to be from the system during training and reintroduce it later as test data. Different techniques split the entire dataset into testing and training subsets using different criteria, with some omitting an entire subject (LOSO validation) while others omit data points across all subjects (k-fold validation). By evaluating a system using unseen test data, more confidence can be had that the system is not overfitting, as the test conditions are more similar to those of real-world deployment.

2.6 . Using Machine Learning Techniques to Detect and Quantify Dysarthric Speech Conditions

Previous literature has demonstrated that machine learning techniques are indeed capable of classifying and quantifying speech quality in both healthy and dysarthric individuals. Discriminating healthy subjects from sufferers of acquired conditions, such as Parkinson's Disease, is a popular area as early detection can lead to much more effective treatment options. Quantifying speech quality objectively mostly stems from the telecoms industry, though the use of telecommunication methods and standards for rating speech have been applied to dysarthric speech to great success. This chapter summarises several examples of previous research and proof-of-concept systems developed to detect and quantify speech dysarthria and associated conditions.

2.6.1 . Classification Algorithms for Discriminating Subjects with PD from Healthy Subjects

Machine Learning systems have been developed to automatically discriminate healthy subjects from people with Parkinson's Disease, with some success.

The features of speech that are extracted and used to train classifiers typically relate to phonation [34]. However, the use of cepstral features like Mel-Frequency Cepstrum Coefficients (MFCCs) has also been explored [8]. Cepstral features of speech are obtained via the inverse-frequency-domain transform of the logarithm of a frequency-domain transformed audio signal. Manipulating an audio signal in this way has the benefit of representing the signal in a way that is more congruent with the human perception of pitch [45]. Phonation features are typically extracted directly from speech recordings using the audio analysis software PRAAT [46].

The system proposed in [1] used a neural network with two dense layers and one hidden layer. The scaled conjugate gradient backpropagation algorithm was used to train the algorithm and the sigmoid function was used at the activation for the hidden layer, while the softmax function was used for the output layer. The database of speech recordings used was the same database created in [47], which is a collection of 45 healthy participants and 45 participants recently diagnosed with PD, speaking the vowel /a/ for 3 to 4 seconds.

The data set was split into a test and training set using k-fold validation, with a k value of 5. Of these folds, 4 were used for training, with the last fold used for testing. The recognition accuracy, sensitivity, and specificity of the system when using only the phonation features was maximal, with a recognition accuracy of 98%, followed by a hybrid of phonation and cepstral features (96.7% accurate), and, finally, purely cepstral features (81.1% accurate).

The results of this study show that, in the early stages of PD, phonation organs like the vocal cords are affected by the disease more noticeably than the speaker's general voice quality, which is what the cepstral features represent.

2.6.2 . Creating Objective Estimates for Subjective Measurements of Dysarthric Speech

The system proposed in [48] provides an estimate of speech intelligibility, expressed as a percentage, using the weighted sum of several objective audio features. The data set for the experiments was obtained from the publicly-available 'Universal Access' database of spastic dysarthric speech and was split into a training set for obtaining the weights and test set for validation [26].

Selected audio features to be assessed were chosen by looking at the common deficits in dysarthric speech such as excessive nasalization, disordered speech prosody, imprecise articulation, and variable speech rate [49]. These features were then individually tested for correlation with the subjective speech intelligibility scores. Using the Spearman rank correlation coefficient and Pearson correlation coefficient, audio features with low correlations were disregarded and the final audio feature set was obtained. The features with the strongest correlations were then used in the weighted sum. The weights used were obtained using monotonic polynomial regression on the training set. To further the accuracy of the system, differently weighted linear mappings were generated based on the severity of the dysarthria.

It has been reported that the features of dysarthric speech change based on the severity of the disorder, which may introduce variance in the correlation of certain objective speech measures with speech intelligibility [50]. To account for the changing influences of prosody or other features as the disorder progresses, the speech recordings were initially classified into low-mid and mid-high intelligibility, with a different set of weightings being trained for each classification.

The methodology used was found to be effective, providing a set of estimates with correlation ratings with the original subjective assessments being as high as 0.97. The highest correlation was indeed achieved when combining a binary classifier for an initial broad categorisation of speech intelligibility and then providing the estimated speech intelligibility score by using a tailored set of weights for each classification.

In a paper that continues the work of [14], the same method of feature correlation and linear mapping can be used in combination with a classifier to produce more accurate results [13]. Rather than classifying people as PD or healthy, a classifier can be used to group PD patients by the severity of their dysarthria. These groups can then have their speech intelligibility assessed using linear mapping functions as in [14], with the main difference being the automatic classification of the groups which further reduces reliance on subjective assessments. In this study, the results of [14] were validated and previously unused glottal speech features were explored. Glottal speech features are heavily reliant on vocal fold vibration, which is affected by dysarthria. These features can be extracted from short-form speech such as individual words, allowing for shorter and easier speech tasks to be undertaken by test subjects who may struggle with longer tasks.

When increasing the feature space used to train an ML system, it is important to consider overfitting. As such, as well as adding glottal features, the proposed system also used a sequential forward selection of features in the hopes of finding the optimal feature subset. A binary SVM classifier was used to label the test data as low-mid and mid-high intelligibility. This classifier was trained using a feature set of 10-15 features, contrary to the composite linear mapping function which contains a maximum of 6 features, as in [13].

Another study attempted to use existing objective measures of speech intelligibility initially conceived for analysing broadcast speech from healthy subjects and apply these measures to Parkinsonian dysarthric speech [15]. These measures were the Itakura-Saito measure (IS), the log-likelihood ratio measure (LLR), and the log-area ratio (LAR). As these techniques are typically used to compare a broadcasting source with a received signal, they rely on the two signals matching exactly in the time domain which is not the case when comparing Parkinsonian speech to healthy speech. To match up the test speech to a golden example of healthy speech, dynamic time warping methods are applied to synchronise the two speech samples. These methods allow for a kind of interpolation using two mapping functions to match up vector representations of the compared speech samples.

The subjective measure used for a baseline comparison was taken by evaluating each spoken sentence from a subject using the GRBAS scale [51]. This is a clinical method of rating hoarse voices, a common feature of Parkinsonian speech. Experimentally, the correlation between IS and GRBAS scores were found to be exceptionally high with a value of 0.8684. Measures of correlation using LLR and LAR scores were much lower at 0.1828 and 0.5142 respectively. The strong performance of the IS measure is thought to be influenced by its consideration of gain as a factor. Parkinsonian voices are often characterised by a significant loss in volume or intensity and this loss in volume can be represented in the IS score [12].

2.6.3 . How Speech Tasks Affect Machine Learning Systems

The study [7] compares and evaluates 18 different feature extraction techniques and 4 machine learning methods to classify data obtained from sustained phonation and speech tasks. Similarly to other literature, the phonation task involved speakers holding the vowel /a/ for several seconds.

All metrics used to assess the classifiers were found to be maximal for the phonation tasks, by a significant margin. The speech task involved the subject speaking a short sentence. This speech recording was then separated into voiced and unvoiced sections using PRAAT [46]. It was found that the unvoiced signal provided marginally greater accuracy than the voiced and complete speech excerpts.

Further research into the performance of different machine learning algorithms and their performance on different speech sources has been undertaken, to determine whether a single classifier should be used across multiple speech tasks [36].

The research described by [36] proposes the use of a different classifier for each type of speech task, rather than one algorithm to classify speech in general. Recordings of sustained phonation, short words and running speech were all used, with feature correlation being calculated for each recording type, rather than over the entire set of speech data. Different classifier algorithms were also tested and used for each recording type, and the final classification decision for a subject is made via a majority vote from all classifiers used.

The feature selection used for each vocal test is chosen by analysing the correlation between each extracted feature with the classification label, using the Pearson correlation coefficient. This, however, leads to the possibility of having a vocal test with no suitable correlated features. In this case, classifications were made by both excluding that vocal test or using the 4 most common features and still attempting classification. It was found that excluding the vocal test led to much more accurate classifications.

The final classification accuracy of the multi-classifier system did show significant improvement over the single-classifier frameworks used in previous literature. An accuracy of 87.5% was achieved by using multiple classifiers and excluding some vocal tests with no correlated features. For comparison, the pre-processed data was used to train several types of classifier and the greatest accuracy achieved was 65.00%.

2.7 . Project Objectives

The background reading presented in this chapter has uncovered previous papers with the same aims as this project. The methods and data presented provide a good model for a successful speech intelligibility estimation system, as well as a standard by which to assess the system created in this project. Research into PD and CP has helped to identify potential novel speech features to extract, in addition to previously used features, to create a new and unique system that hopefully provides an improvement on previous works.

Using this information, the following objectives will be undertaken to fulfil the aims of the project objectives:

- Extract a set of features from the provided speech samples using PRAAT through Parselmouth. The features will be chosen based on their successful use in previous literature and the capabilities of PRAAT to extract novel features pertaining to dysarthria. Once extracted, features will be tested for correlation with subjective speech intelligibility scores.

- Create and train a binary classifier subsystem capable of labelling speech recordings as low-mid and mid-high intelligibility. Several types of classifier will be tested and combined to select a final subsystem with the highest achievable accuracy. The accuracy of the final classification system will be determined using the Leave-One-Subject-Out (LOSO) method.

- Create and train two polynomial regression models to provide an objective estimate of speech intelligibility in recordings. Speech data for all subjects will be split into groups labelled low-mid or mid-high intelligibility, with a separate polynomial regression model being trained on each group. Each of the intelligibility groups will also be split into testing and training sets, utilising cross-validation techniques to properly assess the system. This subsystem will also be tested against a single polynomial regression model evaluating the entire dataset.

3 . The Input Speech Data

In this project, two datasets were used. Originally, the Parkinson's dataset, provided by Prof. Stephen Smith specifically for this project, was set to be used exclusively. However, after initial testing, it was found that the dataset was not diverse enough to properly train any classifier or regression models to achieve reasonable accuracy. Because of this, the UA-Speech dataset [26] was used, a dataset popular in literature for applications such as this.

3.1 . Parkinson's dataset

The Parkinson's dataset is a collection of long-form speech recordings from 72 patients who have been diagnosed with Parkinson's Disease and have undergone DBS surgery. Extensive biographical data and subjective speech assessments made by clinical professionals is also provided for each patient [52]. Each patient produced 4 recordings; 2 of these are from before their surgery where they are on and off mediation, and 2 are from roughly a year after their surgery again both on and off medication.

To produce the recordings, patients were asked to read a short section of text, approximately 30-50 seconds worth per recording. Recordings of their readings were made and used by a clinical professional to produce several subjective and objective measures of their speech quality. These measures include: speech rate (words/minute), intelligibility, and the DAB scale [53], [54]. The included biographical data is incredibly extensive and includes comments on family history of PD, gender, right or left handedness, education and smoking history.

Information is also provided about the DBS surgery each patient has undergone. Each recording is labelled with a date in relation to their surgery and information about the contact placement and anesthesia used is also given.

3.1.1 . Problems With The Dataset

The Parkinsons dataset is incredibly detailed, but when it comes to training a machine learning system it is also limited in some aspects. The majority of participants score very highly in the subjective speech analysis scales used. This is a hindering factor when training an ML system as not enough data on the range of values the system might encounter is provided. This lack of diversity is a common issue in the machine learning research community as systems trained on a homogeneous dataset often struggle when deployed for real-world use, as they have huge gaps in their 'knowledge' which they are often unable to reconcile. This could lead to the system encountering a new data point from a speaker with low intelligibility and being unable to provide a proper and accurate evaluation, having not 'seen' how a human would assess this speaker. Because of this, another dataset was sought out for use in this project.

3.2 . UA-Speech Dataset

The UA-Speech database is a collection of individually recorded words spoken by 20 subjects with cerebral palsy, as well as some biographical data and subjective speech intelligibility scores [26]. Recordings were made using a 7-channel microphone array and a digital video camera, with all set to record simultaneously to different files. All recordings are labelled to denote the speaker, the word spoken in the recording and the microphone used.

For the bulk of the dataset, the Speech material consists of 765 words per speaker, with each word being recorded individually using a prompt. The words chosen were 300 uncommon words taken from digitized children's novels and selected to maximise phone-sequence diversity, 10 digits ('zero' to 'nine'), the 26 phonetic alphabet letters, 19 computer commands (e.g., 'backspace', 'delete', 'enter'), and the 100 most common words in the Brown corpus of written English (e.g., 'it, 'is', 'you'). All words except for the uncommon words were repeated 3 times.

A portion of the dataset was recorded as part of a preliminary study, using the same equipment, the same setup, and a largely similar methodology, except that subjects read a total of only 541 words: digits three times, and radio alphabet letters, computer commands, words from the 'Grandfather Passage', and words from phonetically balanced sentences (TIMIT sentences [18]) one time each. The speakers used in the preliminary study are shown Figure 10, with their data being shown beneath the line.

For each speaker, some biographical data is provided, but it is limited. All participants are American, with English as their native language. The gender and age of each participant are included, though a few of the provided ages are listed only as ‘>18’. Participants represent a spectrum of ages and severities of dysarthria, providing a more complete range of training data, aiding the accuracy of the system created and reducing biases at either end of the intelligibility spectrum. All speakers are diagnosed with some type of cerebral palsy, mostly spastic type, but the diagnosed type for each participant is provided.

Subjective speech intelligibility scores were collected as part of a previous study on the dataset. Five native speakers of American English between the ages of 18–40, with no prior experience with disordered speech, and had no previous training in phonetic transcription were selected to make the assessments. Two hundred words were selected for the test, including 10 digits, 25 radio alphabet letters, 19 computer commands, and 73 words randomly selected from each of the common and uncommon word categories. To ensure accuracy within a listener’s results, 25 words out of the 200 were arbitrarily chosen and uttered twice in the list. Ultimately, an average intra-listener agreement rate of 91.64% was obtained. The final 225 speech files were randomly ordered (with a constraint that repeated words were not adjacent to each other) and presented to the listeners.

Listeners were instructed to provide orthographic transcriptions of each of the 225 words through headphones in a quiet environment, and they were allowed to listen to the words as many times as they liked. Listener transcriptions were then analyzed, and the mean percentage of correct responses averaged across the five listeners was calculated to obtain the subjective intelligibility score of each dysarthric speaker. Based on the averaged intelligibility score, each speaker was then classified into one of four intelligibility categories, namely: very low (0–25%), low (26–50%), mid (51–75%) and high (76–100%). Figure 10 summarizes their demographics.

Speaker	Age	Speech Intelligbility (%)	Dysarthria Diagnosis
M01	>18	very low (10%)	Spastic
M04	>18	very low (2%)	Spastic
M05	21	mid (58%)	Spastic
M06	18	low (39%)	Spastic
M07	58	low (28%)	Spastic
M08	28	currently being rated	Spastic
M09	18	high (86%)	Spastic
M10	21	currently being rated	Mixed
M11	48	mid (62%)	Athetoid
M12	19	currently being rated	Mixed
M13	44	currently being rated	Spastic
M14	40	currently being rated	Spastic
F02	30	low (29%)	Spastic
F03	51	very low (6%)	Spastic
F04	18	mid (62%)	Athetoid
F05	22	high (95%)	Spastic
M01	>18	very low (19%)	Spastic
M02	>18	high (92%)	Spastic
M03	>18	low (29%)	Spastic
F01	>18	low (19%)	Spastic

Figure 10. A summary of the demographics of all speakers in the UA-Speech database [26]

The main focus of this work will be on the participants diagnosed with spastic dysarthria. As discussed in Chapter 1, each type of cerebral palsy presents with its own set of symptoms which would require their own monitoring techniques. As spastic dysarhtria is the most common type and the most frequent diagnosis in the data set, it was the clear choice for use here. In future, it may be possible to extend the system to utilise all recordings in the dataset, and this will be discussed later, in Chapter 10.

4 . Feature Extraction

This chapter describes the feature extraction process for the project. Feature extraction is an important part of the data handling process when developing machine learning algorithms. It allows you to reduce the dimensionality of your data, potentially significantly reducing the training time required as well as the risk of overfitting. Using information gathered through the background research presented previously, this chapter lists the features chosen for extraction, details the process of extracting these features, and reflects on the results.

4.1 . Specification for Feature Extraction Process

1. The first step in specifying the process of extracting features from training data is to decide on a set of features to explore. These features should be objective measurements describing the speech signal which previous literature and research into Spastic Dysarthria seem to indicate are correlated with speech intelligibility.

2. Once these features are identified for testing, the audio dataset should be parsed, with appropriate data structures put into place to collate the extracted feature measurements. In this project, only Python is used for feature extraction, which allows for all of this to occur programmatically.

3. Features can then be extracted from all recordings in the dataset, with raw feature values and z-scores being calculated and stored.

4. Raw feature values are then tested for correlation with the subject's subjective speech intelligibility score. Spearman's Rank Correlation Coefficient and Pearson's Rank Correlation Coefficient will be used for this, as they have been used for this purpose in previous literature [34], [48]. For features seen in previous studies with the same dataset, correlation coefficient values calculated here can be compared to those previously calculated.

5. The top 6 most highly correlated features will then be selected for use in the classification and regression subsystems.

4.2 . Features Chosen for Extraction

The approach for selecting features for extraction is typically two-pronged. Features used successfully in previous literature are, of course, considered. As well as this, new research and novel observations as to the key attributes of the classification results might bring forward new ideas as to potential features which could be used. In this project, features chosen for extraction come largely from previous studies involving the UA-Speech dataset. However, novel features are also introduced to great success. It should be noted that some of these features were used with the intention of being applied to Parkinsonian Dysarthria, as feature extraction research was undertaken early in the project before the UA-Speech was selected to be used. Fortunately, most features selected also correlated with Palsic Dysarthric speech, with the noted exception of jitter. The chosen features are outlined below.

4.2.1 . Prosodic Features

Prosody is the linguistic assessment of speech elements larger than individual phonemes, such as intonation, tone, and rhythmic aspects of speech. Speech prosody is a symptom noticed in all kinds of dysarthric speech and plays a large role in a speaker's overall intelligibility [5], [48], [55]. Three common measures of speech prosody are extracted in this work, as well as two less common features.

Of the more common features, two are related to the fundamental frequency (f0) of the voice, the base frequency which the brain interprets as the pitch of the voice, f0 standard deviation (σf0) and f0 range (Δf0). As dysarthric speech can often be described as monotonous, these measures are expected to correlate strongly [50]. The final common prosodic feature, voiced percentage (%V), quantifies the voiced segments of a speech utterance when compared with the total duration of the utterance. This feature has been identified in previous research as being strongly indicative of disordered speech [56].

As for the less common features, two measures of jitter, subtle and unintentional deviations from the average fundamental frequency, were also selected for extraction. Absolute local jitter is the average absolute difference in frequency, measured in Hz, between two consecutive periods. Local jitter represents the average absolute difference between two consecutive periods, divided by the average period [57]. As one of the dominant symptoms of Parkinson's Disease is muscular tremors,

including muscles affecting the voice, this feature is expected to show a correlation with Parkinsonian speech, but there is little evidence that it will correlate with spastic dysarthric speech.

4.2.2 . Spectral Features

Spectral Tilt is often poorly and inconsistently described in literature, making it difficult to compare studies utilising it as a feature, as differences in implementation can affect the results [58]. Spectral Tilt is most commonly described as a measure of the balance of frequencies in the log-power spectrum. The balance of frequencies in a sound is an important feature in speech as it potentially encompasses articulation, intonation, vocal stress, lexical stress and vocal quality in one measure [59]. In this paper, the implementation of spectral tilt is calculated using the average gradient of the log-power spectrum.

Harmonic-to-noise ratio (HNR) is a quantitative measure of the additive noise in a vocal signal. Additive noise is usually a result of turbulent airflow from the glottis, caused by inadequate closing of the vocal folds. Aperiodic vibration of the vocal folds is also a cause of additive noise. HNR is thus a measure of the dominance of periodic over aperiodic levels in the vocal signal [60]. Vocal utterances with lower HNR values are often described as breathy or strained, much like the voices of dysarthric speakers. This indicates the potential for HNR to be a strong correlate of intelligibility in dysarthric speech.

4.2.3 . Cepstral Features

Cepstral analysis is a tool for investigating periodic structures within frequency spectra, with the power cepstrum being particularly important for analysing human speech. The zeroth-order cepstral coefficient (*C0*) denotes a measure of short-term log-spectral energy, with the delta zeroth-order cepstral coefficient (*ΔC0*) denoting the rate of change of this log-spectral energy.

$$\Delta c_0(m) = \sum_{l=-L}^{L} l \; c_0(m+l) \qquad (2)$$

ΔC0 is calculated as above, with *L=3* being used, as in [34]**.** To calculate the cepstral coefficients, the input audio signals are framed into 30ms windows in order to sufficiently capture phoneme information. A pre-emphasis filter with a value of 0.97 is also applied to the signal. The standard process of generating cepstral coefficients is outlined in Figure 11 below [8].

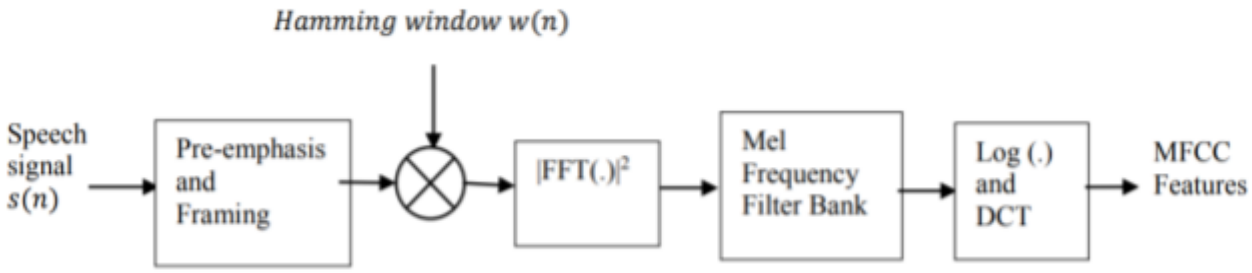

Figure 11. The algorithm for calculating cepstral coefficients from a speech signal [8].

4.2.4 . Feature Standardisation

Standardisation in this concept is the process of scaling measurements to be zero-mean with unit variance. This process is often carried out when preparing data to be passed to a machine learning system when the extracted features vary in their ranges or units of measurement. This is true for the features chosen in this paper.

Standardisation is an important process for machine learning when using classification or regression models which place significance on distance computations. When using a range of features, if the range of one feature is significantly larger than others, that feature can dominate the performance of an ML system, even if it demonstrates a weak correlation with the desired output.

Z-score is a popular implementation of data standardisation in machine learning systems and is calculated with equation 3. As several distance-based, clustering and regression algorithms are being used in this project, the z-score will be utilised in place of the raw feature values.

$$z = \frac{value - mean}{standard\ deviation} \tag{3}$$

Something to note when standardising data is that when novel data is presented to the system and feature extraction takes place, the z-score will be calculated for the novel data based on the cached mean and standard deviation from the training data. This places importance on the initial training data providing a properly representative mean and standard deviation.

4.2.5 . Feature Selection and overfitting

Once all the features have been extracted and tested for correlation, the most highly correlated features will be selected for use. It is possible to achieve a seemingly better performing model when using many features, regardless of correlation, but this can cause the system to overfit.

By pre-assessing features for correlation with the subjective speech analysis, a smaller set of features can be used to maximum effectiveness. This ensures that the final system is actually being trained to recognise the classes, rather than simply sorting the provided recordings.

4.2.6 . Implementation of Feature Value Calculation

With the set of exploratory features decided upon, methods extraction must first be implemented for each. A popular piece of software for vocal analysis is PRAAT [46]. PRAAT does feature batch processing scripts for extracting features from many recordings but scripting is limited and any post-processing must then be carried out externally. To create a more cohesive workflow, one which will work especially well later with the machine learning workflow, a Python library called Parselmouth is used to bind PRAAT processes to Python commands [61].

4.3 . Method

The process of feature extraction begins as an exercise in file management and file system crawling, as the recordings must be gathered and sorted before they can be processed in a meaningful way. The UA Speech database separates recordings by participant and labels each file with the participant designation, utterance designation, and microphone channel used. To ingest these files and keep the metadata, several Python classes were created.

- **Participant**. This class represents one participant in the study and will store all the relevant information for a participant including their designation and subjective intelligibility score.
- **Recording**. This class represents one recording made by a participant. It has properties for storing the filename of the recording and the feature values. The feature value properties are initialised as null and only contain values once the feature extraction method is called and finishes successfully.

As each participant has their own folder within the database, each folder can be programmatically accessed, with an instance of the *Participant* class created for each folder. Each participant folder can then be crawled and an instance of *Recording* can be created for each .wav file within, where the name of the file contains 'M6'. This check ensures only files from microphone 6 are used and no utterances are duplicated by using recordings from multiple microphones.

At this stage, no features are immediately extracted. This first step provides a labelled reference to the codebase for each relevant recording within the dataset. Feature extraction is a computationally expensive process, so minimising the duplication of tasks is important. Feature extraction is triggered within a single script, which iterates over every instance of the Recording class, calls the method to extract the features, and stores the values for subsequent processing and analysis.

It is important to note that only the feature values are extracted and stored; the z-score for each feature is calculated only when it is used. The reason for this is LOSO cross-validation. As previously stated, when the system is deployed under real-world conditions, the new data will be standardised using the cached mean and standard deviation from previous training data. To replicate this, when using LOSO cross-validation, the test data is also left out of the z-score calculation. Hence, this is calculated at run time and not stored. To calculate the correlation between each feature and the participant's subjective speech intelligibility, the raw feature values are used. The z-score is used later for classification and regression.

4.4 . Feature Correlation Results

Table 1 shows the results of correlation analysis between the standardised extracted features from each speech recording and the subjective intelligibility score of the corresponding subject. The Spearman's Rank Correlation Coefficient (Rs) and Pearson's Rank Correlation Coefficient (Rp) are used for correlation analysis and the results of the same analysis undertaken in [34] are also shown in the table where applicable (Fs and Fp respectively).

The 6 most highly correlated features are shown in bold. These are the features ultimately chosen for use within the system. Also shown in the table are the correlation scores for the same features across the same dataset reported in [48].

Feature	Rs	Rp	Fs	Fp
Fundamental standard deviation	**-0.23**	**-0.18**	**-0.43**	**-0.57**
Fundamental range	**-0.30**	**-0.28**	**-0.72**	**-0.76**
Voiced percentage	**-0.44**	**-0.40**	**-0.75**	**-0.77**
Delta C0 standard deviation	**0.29**	**0.12**	**0.81**	**0.71**
Spectral tilt	**-0.20**	**-0.13**	**N/A**	**N/A**
Harmonic noise ratio	**-0.35**	**-0.33**	**N/A**	**N/A**
Local jitter	0.01	-0.04	N/A	N/A
Local absolute jitter	0.16	0.15	N/A	N/A

Table 1. The Spearman's (s) and Pearson's Rank (p) correlation values for the features extracted in this work (R), as well as the scores observed by [48] (F)

4.4.1 . Analysis of the Results

The 8 features tested can be split into 2 categories; features from literature, and novel features. As expected, the features from literature all performed well, though the correlation is lower than expected. These features have been previously used in successful classification systems and their use is strongly backed up by research into spastic dysarthria.

Of the novel features, local jitter and local absolute jitter performed poorly, though this was expected. Jitter or micro-oscillations and tremors in the voice is a symptom typically of Parkinsonian Dysarthria, not Spastic Dysarthria. One reason for testing it at all was because it had already been implemented for use with a previous dataset; it was not added specifically for use with Spastic Dysarthria. However, due to the anomalously low scores of some features, local absolute jitter does come close and certainly should be considered in future work.

The Spectral features, Spectral Tilt and Harmonic Noise Ratio (HNR) performed very well, with HNR providing the second-highest correlations seen. While these features are not common in previous implementations of classifiers for dysarthric speech, they have been previously used in studies of voice pathology, so their success is not unexpected [62]–[64].

4.4.2 . Problems with the Correlation Results

An interesting and confusing finding from the correlation results was the severely diminished correlations in the results found here when compared to previous literature. Most of the results from this study sit at around half the expected value. Reviewing the previous study in question reveals that they should be using the same data and processes, with one exception [48]. While it is not made explicitly clear, the previous study may have been correlating the extracted features with individually labelled recordings, rather than labelling each recording simply with the averaged intelligibility attributed to the subject. This would mean that in cases where subjects found certain words easier to pronounce than others, varying the intelligibility of a single recording in regards to the mean, the mean subject intelligibility would stray from the individual label. This could account for the discrepancy in correlation scores between the results presented here versus previously reported results.

5 . Classification System

This chapter describes the implementation of the classification subsystem. The requirements of the first subsystem are to create a binary classifier that can reliably group speech recordings by their intelligibility. The two labelled groups of recordings will later be used to train two instances of the second subsystem, the polynomial regression model. This chapter will cover the development of the classification subsystem through the implementation and testing of 9 types of binary classifier, as well as the combination of these classifiers.

5.1 . Specification for the Classification Subsystem

1. Each of the classifiers chosen for use will first be implemented using Scikit-Learn [65]. They can then be trained and tuned for optimal individual performance. This will provide the baseline for testing the comparative performance of ensemble algorithms.

2. All ensemble algorithms can then be created and evaluated using the output from all optimally-trained individual classifiers. At this stage, the performance of ensemble algorithms can be analysed, compared to individual classifier performance, and a classification method can be identified as the primary candidate for use in this subsystem.

3. The identified classifier candidate is now selected and the regression subsystem can be developed. After the regression system is developed, a final analysis of the entire system, including a more in-depth analysis of the regression system, will be performed and final subsystem results will be collected and analysed.

5.2 . Classification Labels

In previous works, the two labels used for classification are: subjects with *low-mid* intelligibility (0-50%) and *mid-high* intelligibility (51-100%) [34], [48]. The system implemented here uses the same labels but the boundary point of 50% will be varied experimentally to hopefully improve classification accuracy and the accuracy of the objective intelligibility measure.

However, the movement of the label boundary will potentially affect both classification accuracy and regression accuracy. The accuracy of the regression subsystem is the priority, as it produces the final user-facing output; the classification label is used internally only. For this reason, the classification label boundary test will be made later, on the completed system. While testing the classification subsystem only, the label boundary will remain fixed at 50%.

5.3 . Evaluation Methods

Choosing an evaluation method for machine learning systems is incredibly important in confidently analysing the performance of the system. Utilising different metrics and data-splitting techniques, it's incredibly easy to misrepresent the performance of a system, leading to widespread misinformation in the media and scientific communities, as well as adversely affecting the wider world when a misunderstood system is deployed [66], [67].

During this project, two techniques for evaluating Machine Learning (ML) systems are utilised. Both of these techniques present different advantages and disadvantages, with each being used previously in works involving the same data [34], [48].

5.3.1 . Leave-One-Subject-Out

Splitting a dataset into a training and a test set using Leave-One-Subject-Out (LOSO) cross-validation involves removing every recording made by one subject, with the removed recordings set aside as the test set. One of the main advantages of this, in this context, is that when it comes to validating the ML model, it has never seen data from this subject before, which reduces the risk of the model overfitting. If the model has seen data from the subject before, there is a risk of the model being able to correctly classify their data by relying on prior knowledge of an aspect of their data not related to the overall goal of the classifier. Accuracy results from a system validated with the LOSO method are, hence, considered more reliable than using data from all subjects for testing and training, and are less prone to false inflation.

This does come at a cost, however. To properly validate a system, it should be trained and tested once each with every subject removed. So for *N* subjects, the model is trained and validated *N* times. The final assessment of the system is made by taking the mean average of results for individual runs, providing a singular final result that describes the performance of the system with any one subject missing.

This can be a very lengthy and computationally expensive process. The dataset used in this study consists of 11 subjects with over 700 voice recordings per subject. With an average training and validation time of around 30 minutes, a relatively short time in the world of Machine Learning, it would take around 7.5 hours to evaluate the accuracy of the system after any changes were made. This makes it, of course, impractical to explore a range of changes to a system to increase its accuracy. For this reason another, less intensive, validation technique was also employed in some areas.

5.4.1 . Common-Uncommon Split

Utilised in previous works involving the Universal Access Speech (UA-Speech) database, the Common-Uncommon Split method leverages the fact that the database contains labelled sets of recordings using several word lists, with some being composed of common everyday words and one list being uncommon phone-diverse words. Dividing up the speech data in this way is similar to k-fold validation, perhaps the most popular machine learning model data partitioning and validation technique. While the idea might be similar to LOSO, a portion of the overall dataset is unseen by the model during training and used only for testing, the training set is much smaller. This makes the training process much faster. In the UA-Speech database, common words, digits, alphabet letters, and computer commands make up just 61.79% of the database with the remaining 39.21% being uncommon words. Comparing this 60/40 training-testing split to the 93/7 split when using LOSO clearly shows the huge disparity in computational intensity between the two methods.

With such a large disparity, it is easy to doubt the effectiveness of the common-uncommon split method. The content of the uncommon word list, however, is not particularly comparable to the common word list. The phonetic diversity of the uncommon word list is intentionally maximized to cover a wide spectrum of potential utterances. That being said, there is still potential for bias when using this validation method. To reduce this bias or its effect on the final system, the common-uncommon split will only be used for tuning hyperparameters and not for the final validation of the model. This way, the decreased training time of the system can be leveraged to evaluate tweaks to the model while the accuracy of the LOSO method will provide more robust results for publishing.

5.4 . Method

Inspired by previous literature, which implements several classifiers to provide more reliable labelling through repetition and consensus of those classifiers, this project also takes a similar approach [36]. Previously, it was suggested that different types of classifiers should be used on different speech tasks to improve accuracy through a majority vote of these classifiers. The dataset used in this work consists of only one speech task. However, the idea of using several classifiers and a majority vote to improve overall accuracy is one worth exploring here.

5.4.1 . Novel Ensemble Algorithm

In this project, all 9 of the classifiers previously described in Chapter 2.4 were independently implemented, trained, and tuned in Python using Scikit-Learn [65]. As well as evaluating the performance of each classifier, the classifiers were then combined to create ensemble algorithms. The performance of each ensemble algorithm and traditional classifier are then compared.

Each classifier first predicts every item in the test data. These predictions can then be analysed to evaluate the performance of each classifier using common metrics such as accuracy, specificity, and confusion matrices. Every possible combination of n classifiers, where n is an odd number between 3 and 9 inclusive, is then calculated, and the predictions for each combination of classifiers are collated to form a subset of all predictions made. For each combination of predictions, a majority vote is taken for each recording. The result of this majority vote forms the set of predictions for each combination of classifiers. This set of predictions can then be evaluated in the same way as prediction sets for individual classifiers. Figure 12 below shows an example of this ensemble algorithm for 2 recordings and 4 classifiers.

For each recording, individual predictions are made

This table shows 4 classifiers with accuracies of 100%, 50%, 0% and 50% respectively

Recording	Actual Label	SVM	GP	KNN	DT
Subject 3, UW4	*Low*	*Low*	*Low*	*High*	*High*
Subject 3, UW5	*Low*	*Low*	*High*	*High*	*Low*

With 4 classifiers, 4 combinations of size r=3 can be made

For each combination, the majority vote of the classifier set is taken as the result

SVM, GP and DT combined show an accuracy of 100% for the two recordings used

Ensemble	Ensemble Result (UW4)	Ensemble Result (UW5)
SVM, GP, KNN	*Low*	*High*
SVM, GP, DT	***Low***	***Low***
SVM, KNN, DT	*High*	*Low*
GP, KNN, DT	*High*	*High*

Figure 12. An outline of the process for creating an ensemble classification algorithm from 3 classifiers

Odd numbers of classifiers are used exclusively, to avoid the need for a tie-breaker function. Implementing a tie-breaker function could be an area for future work, allowing for twice the number of classifier combinations to be tested. However, if not implemented carefully, there is potential for bias and inaccuracy to be introduced. To implement this properly would require further research outside the proposed scope of this project.

5.4.2 . Initial Classifier Training and Hyperparameter Tuning

Before the classification algorithm is formally trained and assessed, and before combinations of classifiers can be assessed, hyper-parameters for each classifier must undergo some tuning to optimise their performance. Hyperparameter tuning is a vast subject area so, to keep the project to a reasonable scope, hyper-parameter values are varied within a specified range, using workflows implemented in Scikit-Learn.

The hyperparameter tuning workflow first initialises each classifier with default values. Classifiers are then trained and evaluated using the common-uncommon split cross-validation technique described previously. This allows for the tuning process, which requires training and retraining classifiers many times, to operate in a way that is less accurate but far less computationally intensive. Relevant parameter values are increased and decreased incrementally based on their effect on the classifier accuracy, covering a small area within the parameter space. This process results typically in minor accuracy increases, within a few percent. Many specialized hyper-parameter tuning algorithms exist and may result in further improvements at the expense of more computational complexity, providing scope for future improvements. The results of this hyper-parameter tuning are shown in Table 2.

classifier	Parameters changed	Accuracy change from default
SVC	Gamma = 1.7, C=0.8	+2%
MLP	alpha=1, max_iter=3000	+0.5%
DT	max_depth=5	+0.7%
GP	0.8*RBF(0.8), n_jobs=-1	+1.2%
KN	k=3	+0.3%
RF	max_depth=7, n_estimators=10, max_features=2	+0.9%
ADA	n/a	+0%
GNB	n/a	+0%
QD	n/a	+0%

Table 2. The optimal hyper-parameter found through manual search and their effect on the performance of each classifier

As the table shows, the hyper-parameter was successful, improving the accuracy of the support-vector machine classifier by 2%. Several classifiers remained unchanged as the default values were already found to be optimal, within the parameter space which was explored here. There are several important things to note here, however.

Firstly, it is not entirely accurate to rate the performance of a classifier with one number. Receiver Operating Curves (ROC) and Area Under the Curve (AUC) values are typically used to give a more complete picture of the change in performance of a system as the parameters are tuned. This allows for data scientists to look at how the false-positive and negative rates change over time, as well as the overall performance. As the approach taken here uses a simplified version of the typically more extensive hyper-parameter tuning workflow, it was considered acceptable to just optimise for maximal accuracy.

5.4.3 . Classification Subsystem Training for Ensemble Algorithm Formation

With each classifier now performing optimally on its own, tests can be conducted to determine the most accurate classifier or combination of classifiers. By combining the predictions each classifier made with its most optimal parameters, 247 ensemble algorithms can be formed by generating all combinations of n = 9 where r is an odd number from 3 to 9 inclusive, as well as the 9 solo classifiers. To evaluate the classifiers under conditions closer to those of real deployment, providing the most accurate results, LOSO cross-validation was used. The final accuracy value was calculated by taking the mean value of the accuracy of the system after each run. After 15 runs, one with each subject removed, the 5 most accurate classifiers or combinations of classifiers were found and are shown in Table 3.

Classifier(s) Used	Accuracy (%)
'GP'	79.72
'MLP', 'GP', 'KN'	79.24
'MLP', 'DT', 'GP', 'KN', 'ADA'	79.13
'SVC', 'MLP', 'GP', 'KN', 'QD'	78.94
'SVC', 'MLP', 'DT', 'GP', 'KN'	78.68

Table 3. The best performing classifiers and ensemble algorithms

The results of this test show that the Gaussian Process Classifier, a solo classifier, is the best performer by a very small margin. With the margin being so small, it is possible that an ensemble algorithm could be shown to be the most accurate if a larger dataset were to be used, especially since the next 4 most accurate algorithms are ensemble algorithms that utilise the Gaussian Process Classifier within. It would only take a very small number of cases in which the Gaussian Process Classifier was mistaken and overruled to have one of the ensemble algorithms perform the best. Furthermore, under ROC and AUC analysis, it is possible that ensemble algorithms showed less confusion and would be the more optimal choice in some situations.

5.4.4 . Gaussian Process Classifier Performance Analysis

Going forward from this point, the solo Gaussian Process Classifier was chosen for use in the classification subsystem. To better understand the performance of the classifier, more in-depth results and analysis are given below in Table 4.

	Precision	Sensitivity	F1-Score
mid_high	0.85	0.75	0.80
low_mid	0.75	0.85	0.80
accuracy			0.80

Table 4. A more detailed look into the performance of the most accurate classifier, the Gaussian Process classifier

From the precision and sensitivity across the mid_high and low_mid labels, we can see that the performance of the system is quite balanced, with a slight performance increase for *low_mid* severity. With the current parameters, misclassification either way is roughly equally likely. In a non-critical application like this, where the system output is not expected to directly diagnose a patient, this is a perfectly acceptable balance. If the classification were to be used in future for directly diagnosing dysarthria or associated conditions, it might be preferable to tweak the system parameters to increase sensitivity for mid_high severity. This would introduce more false positives but fewer false negatives, potentially wasting more of a clinician's time through erroneous referrals but ultimately finding more true positives, which results in more accurate classification of more severe patients.

6 . Speech Intelligibility Estimation System

This chapter describes the creating and testing of the speech intelligibility estimation subsystem. Two systems will be created and compared; a single-stage system relying on just the standardised extracted feature values and a two-stage system that also utilises the estimated intelligibility band label provided by the classification subsystem. Each will utilise a polynomial regression model or models which will map the feature values to an objective estimate for the intelligibility of the associated utterance.

6.1 . Specification for the Speech Intelligibility Estimation System

1. Polynomial regression will be implemented in Python using Scikit-Learn [65]. It is not natively supported by Scikit-Learn, so some implementation is necessary. The model will then be trained using the same standardised features provided to the classification subsystem.

2. The degree of the polynomial regression model will then be determined by training the model with a degree of 1, 2 and 3. As previously discussed, degrees higher than this are typically not used as they are especially prone to overfitting, and computational complexity increases exponentially. The model with the lowest r-square error will then be chosen as the baseline for the performance of a single polynomial regression model across the whole dataset.

3. Once a single polynomial regression model is created, trained and analysed, two more will be created. The test data will be split into *low_mid* and *mid_high* intelligibility utterances. Each polynomial regression data will then be trained using only one set of utterances. The performance of the models will then be analysed separately before their combined predictions are analysed and compared to those from the single regression model.

6.2 . Cross-Validation

Ideally, LOSO cross-validation would be used for training and testing the system. However, in the case of using two regression models, one for utterances from *low_mid* and *mid_high* intelligibility speakers, the dataset for each model is incredibly small. Using 11 speakers to represent all spastic dysarthric speakers in the world is naive, but due to the nature of studies like this, it is accepted. Dividing this dataset into two groups and then removing another subject from each dataset creates far too small of a training set to be meaningful. LOSO would be usable on the single regression system, but then the results would be incomparable to the proposed dual regression system. Due to this, the common-uncommon split will be used to cross-validate the regression systems.

6.3 . Averaging Predictions

The ultimate goal for this system is to provide an objective measure of the intelligibility of a speaker, not just to rate single recordings. To that end, previous studies in this area involving the same dataset have provided plots in which a single prediction is made for a speaker [34], [48].

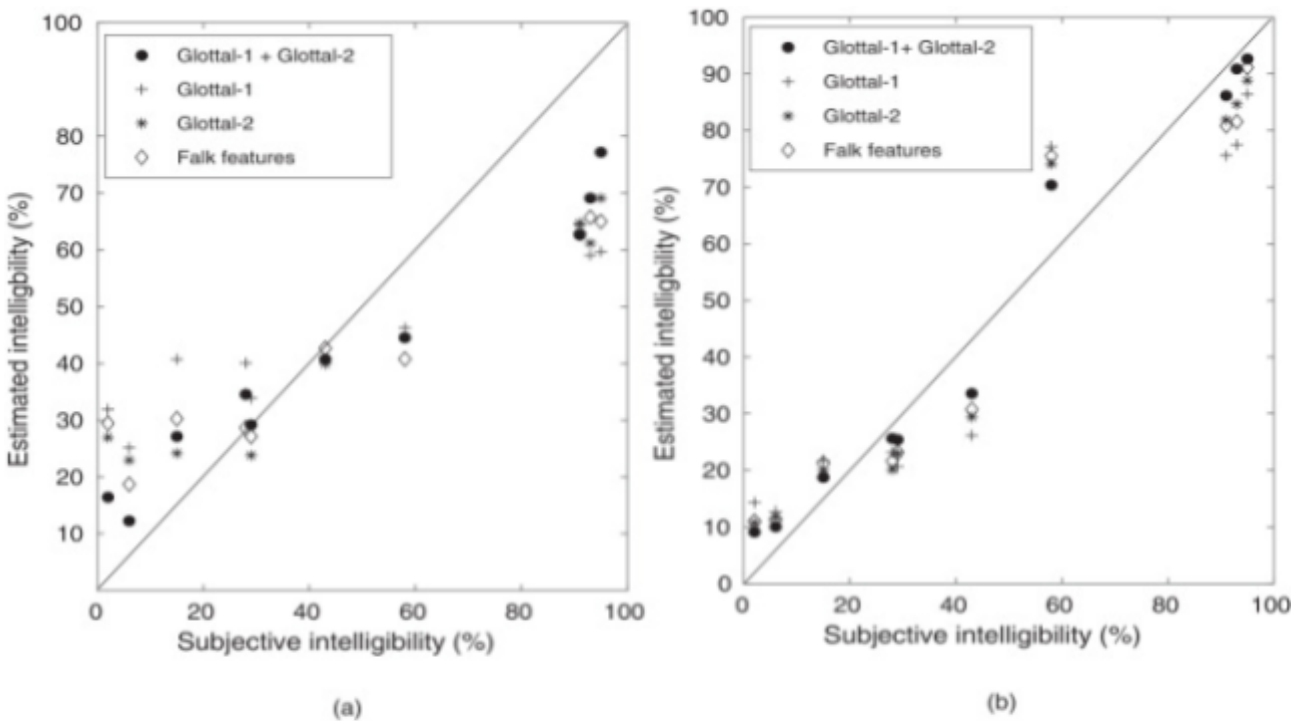

Figure 13. Estimated intelligibility versus subjective intelligibility obtained for (a) a single-stage and (b) a two-stage intelligibility assessment system developed in [48].

Without error margin indicators, it is difficult to get the full picture, as potentially hundreds of recordings are averaged to produce a single mean average value for a speaker and it is unclear exactly how many values are averaged to produce the plots above. This mean average might correlate highly with the subjective results desired, but real systems are unlikely to be used if they require the subject to provide several hundred recordings before being provided with a result. However, root-mean-square-error values are provided later in the paper which, in effect, provides suitable information in place of plotted error bars.

6.4 . Method

The single-stage system was developed first, providing a baseline for the performance of the two-stage system. This will provide results to test the hypothesis that dysarthria symptoms present in different quantities based on the severity of the causing condition, specifically spastic type cerebral palsy [48], [68]. The implementation performing the most favourably in terms of R^2 and RMSE error will be selected for use as the final system.

6.4.1 . Single-Stage Subsystem

The single-stage Regression system utilises features extracted from every utterance across all speakers, specifically the z-score of each feature is used. These features are then split into a training and test set using the common-uncommon split method described in the previous chapter. The features can then be expanded to a polynomial using the method previously described, with degrees of 1, 2 and 3 each being tested. The performance of the single-stage system is described below in Table 5.

Degree	R^2 Error (%)	RMSE Error	Rs	Rp
1	49.87	24.96	0.85	0.89
2	57.42	23.00	**0.91**	**0.93**
3	**63.57**	**21.28**	**0.91**	**0.93**

Table 5. A performance comparison of the single stage regression system with various degrees of polynomials

It's clear from the table that even when performing at its best, fitting a 3rd-degree polynomial, the performance of a single-stage regression system is far too poor to be usable. The correlation coefficients do show that there is a fairly strong trend between the subjective intelligibility scores and the predictions from the model, but the RMSE error score shows that the predicted scores are frequently off by upwards of 20%. The level of variance in the objective scores are far too high to be usable.

The model does improve as the degree of the polynomial increases, indicating that the relationship between the features and the subjective intelligibility results is nonlinear. These improvements are somewhat minor however and further increments to the degree of the polynomial would be at risk of overfitting, as the model becomes more flexible and begins to tightly adhere to individual training points.

6.4.2 . Polynomial Interpolation

Polynomial regression can be implemented in Scikit-Learn using a linear regression model and expanding the features to produce a polynomial. Using the process of polynomial interpolation, a matrix of pseudo-features is created from the original feature set and the new weights generated are optimised through a standard linear model to produce a result that will fit nonlinear data. A typical linear regression model with two features might look like the following.

$$\hat{y}(w, x) = w_0 + w_1 x_1 + w_2 x_2 \tag{4.1}$$

Three weights characterise the intercept (w_0) and the influence of each feature on the model (w_1, w_2). To fit these weights to a polynomial through interpolation, the equation can be expanded to the appropriate degree. For the case of a 2nd-degree polynomial, expanded from the linear equation above, the following equation would be produced.

$$\hat{y}(w, x) = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_1 x_2 + w_4 x_1^2 + w_5 x_2^2 \tag{4.2}$$

The weights associated with the new pseudo-features are optimised through the linear regression algorithm already implemented into Scikit-Learn. This provides a polynomial model which is trained leveraging the fast performance of linear regression, with the additional weights affecting performance linearly.

6.4.3 . Dual Regression Model Training and Evaluation

The two-stage system consists of two polynomial regression models, each created and trained as described for the single-stage system. The one key difference is that rather than being trained and validated using the entire dataset, each model is trained on just utterances from *low_mid* or *mid_high* intelligibility speakers. In a deployed system, the automatic classification system described in the previous chapter would first classify a recording before passing it through the appropriate regression model, hopefully providing a more accurate objective intelligibility score than a single-system approach.

Each model is created, trained and has its degree optimised separately. Once each model is optimised and trained, predictions are made for every utterance in the data set, using the appropriate model, and these results will be compared against the single-stage baseline subsystem.

Degree	R^2 Error (%)	RMSE Error	Rs	Rp
1	26.53	9.69	0.90	0.74
2	53.20	7.74	0.90	0.89
3	**59.28**	**7.22**	**1.00**	**0.93**

Table 6. A performance comparison of the *low_mid* regression model with various degrees of polynomials

Degree	R^2 Error (%)	RMSE Error	Rs	Rp
1	69.76	10.90	0.54	0.92
2	83.28	8.10	0.77	**0.98**
3	**86.31**	**7.33**	**0.96**	**0.98**

Table 7. A performance comparison of the *mid_high* regression model with various degrees of polynomials

Performance of the individual regression models which make up the two-stage system present a substantial improvement over the single-stage system in almost all metrics, the most drastic of which being RMSE. The RMSE scores of the 3rd-degree polynomial regression models for *low_mid* or *mid_high* intelligibility have dropped to single-digits, performing comparably to previous literature [34]. The Spearman's Rank Correlation Coefficient for the *mid_high* intelligibility system does present an outlier, with the correlation seeming to be lowest while every other metric is maximal. Ignoring this outlier, it's clear that the subsystem going forward should utilise 3rd-degree polynomial models in both instances for the best performance.

6.4.4 . Two-Stage Subsystem Results

With both regression models making up the dual regression subsystem trained and optimised, they can be assembled into the final subsystem for comparison against the single-stage subsystem. The performance will be evaluated bypassing each validation utterance, using the common-uncommon split, to the appropriate model and analysing the results as a whole, as if the predictions were provided by a single model. It's important to note that for these purposes, it is assumed that every utterance is always correctly classified and passed to the appropriate model by the previous subsystem.

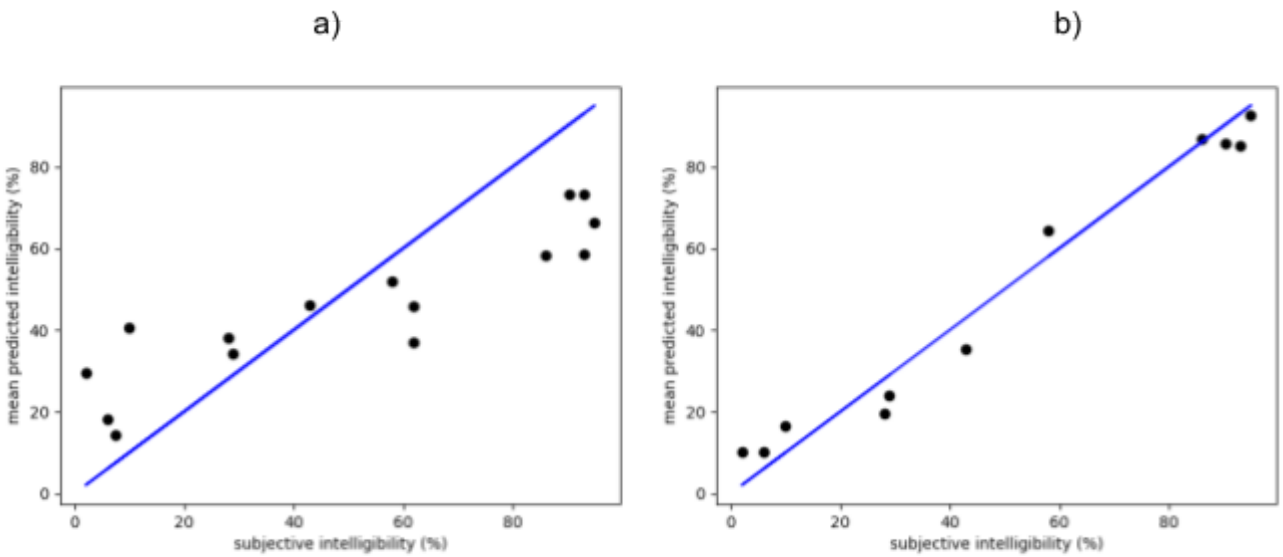

Figure 14. Estimated intelligibility versus subjective intelligibility obtained for (a) a novel single-stage and (b) two-stage intelligibility assessment system.

Figure 14 above shows the mean average of regression predictions for the case of a single-stage system vs a dual-stage system. A data point on the line $y = x$ (shown in blue) indicates a prediction that exactly matches the subjective score. It's immediately clear that the dual-regression system (chart b) provides estimates which are far more accurate, producing a cloud of data points following the line $y = x$. This echoes the results of previous work and falls in line with the hypothesis presented in previous literature that the symptoms of spastic dysarthria present in different degrees depending on the severity of the case [34], [48], [68].

	R^2 Error (%)	RMSE Error	Rs	Rp
Single-Stage	63.57	21.28	0.91	0.93
Two-Stage	**95.73**	**7.28**	**0.96**	**0.98**

Table 8. A comparison of the novel single-stage and two-stage systems developed in this paper

The performance indicators shown in Table 8 show just how much of an improvement is gained by splitting the data by severity. Every metric is improved by moving to a two-stage approach with no apparent compromises. RMSE error is now reduced to just 6.12, a far cry from the unacceptable error of 42.66 seen previously. To put this in the context for comparison to human listeners, the average intra-listener agreement rate from the listeners providing the subjective analysis was 91.64% [26]. Comparing that to the R^2 and RMSE error values from the two-stage system, it's clear that it's not unreasonable to see a system like this being deployed to a clinical setting to replace human subjective speech analysis. Though without access to the individual scores, it is difficult to directly compare regression errors to human error.

7 . System Assembly and Evaluation

Both potential implementations of the speech intelligibility system have been compared and the performance of the two-stage system is found to be demonstrably far superior to the single-stage approach. The two-stage system is not entirely complete at this stage, however. Further exploration to potentially improve the performance of the system is to move the boundary between *low_mid* and *mid_high* intelligibility labels. Furthermore, there is the question of what happens when the first subsystem misclassifies an utterance. This is a problem unique to the two-stage approach and is not answered by previous literature.

7.1 . The Label Boundary

The decision in previous works to split the groups at 50% was originally decided to make it straightforward for an inexperienced speech therapist to label recordings [48]. In later works building upon this, the groups were kept the same rather arbitrarily, even though machine learning methods were implemented to label recordings rather than a human. In a finished system, the user would be presented with an objective estimate of their speech intelligibility and not see the classification label applied to their speech recordings; the label is purely used internally. Furthermore, no previous research has been done into the effect of moving this boundary on classification accuracy and the accuracy of the objective intelligibility estimate. As mentioned previously, it is known that different severities of dysarthria present different symptoms and in different quantities [3]. However, the exact boundary between high and low intelligibility speech is not explored. The experiment here aims to explore the effect of moving this boundary on the overall system accuracy.

The boundary between the low-mid and mid-high intelligibility labels was varied from 25% up to 60%. Both the classifier and regression subsystems were trained and evaluated separately using LOSO validation and the common-uncommon split respectively at each label boundary level, changing the percentage of recordings falling into each category as the subjects are re-labelled to meet the new boundaries. The results of this are shown in Table 9.

Boundary (%)	Classifier Accuracy (%)	Regression R-Squared Error (%)	RMSE Error	Rs	Rp
25	84.78	85.48	12.16	0.96	0.96
30	81.84	97.00	6.11	0.96	0.99
35	81.84	97.00	6.11	0.96	0.99
40	**81.84**	**97.00**	**6.11**	**0.96**	**0.99**
45	79.72	95.73	7.28	0.96	0.98
50	79.72	95.73	7.28	0.96	0.98
55	79.72	95.73	7.28	0.96	0.98
60	80.28	95.62	7.13	0.99	0.98

Table 9. Results from varying the intelligibility classification boundary and how classifier accuracy and regression errors are affected

From the table, there is not a boundary percentage that presents as the clear winner. Many of the boundary percentages show no difference in the final results at all. One reason for this might be that there are fewer data points in this range. Of the 11 subjects in this study, 4 have subjective intelligibility scores over 60%, and 3 have scores below 25%. With over 63% of subjects lying outside the experimental range, changes to the expected *low-mid* and *mid-high* groupings throughout the tests are minimal. Given a larger and more varied dataset, it is possible that a larger difference across varying boundary percentages might present itself. However, the effect demonstrated here is minimal.

A final boundary percentage of 40% was selected for the final system as it shows a marginal improvement in classification accuracy, r-squared error rate and Pearson's rank correlation over a boundary of 45% and 50%. There is an argument to be made that a boundary of 25% should be chosen, as it boasts the best classification accuracy, a large increase of 2.94% over a boundary of 30-40%. However, this comes at the cost of a far lower r-squared error rate and far greater rmse error for the objective intelligibility estimate that is, after all, the final output and true purpose of the system.

7.2 . Regression Error for Misclassified Utterances

A unique potential problem for using a two-stage system is the potential for the classification stage to mislabel an utterance, passing it to a regression model whose weights are potentially not tuned to provide an accurate intelligibility estimate. This is not covered in previous literature; previous studies have provided their system results based on accurately labelled data [34], [48]. The accuracy of the classification subsystem is 81.84%, as it is currently configured. False positives and false negatives occur at approximately the same rate. This means that roughly a fifth of all recordings are potentially mislabelled.

To assess the effect of misclassification of an utterance on the accuracy of the objective regression score for that utterance, the following experiment was conducted. The regression system developed and trained previously was given the validation data, just as in the previous tests of the dual-regression model. However, the *low_mid* and *mid_high* labels applied to the data were inverted, so the *mid_high* regression model was providing estimates for *low_mid* utterances. The chart below shows the mean objective intelligibility estimate for each speaker calculated by the inverted regression models.

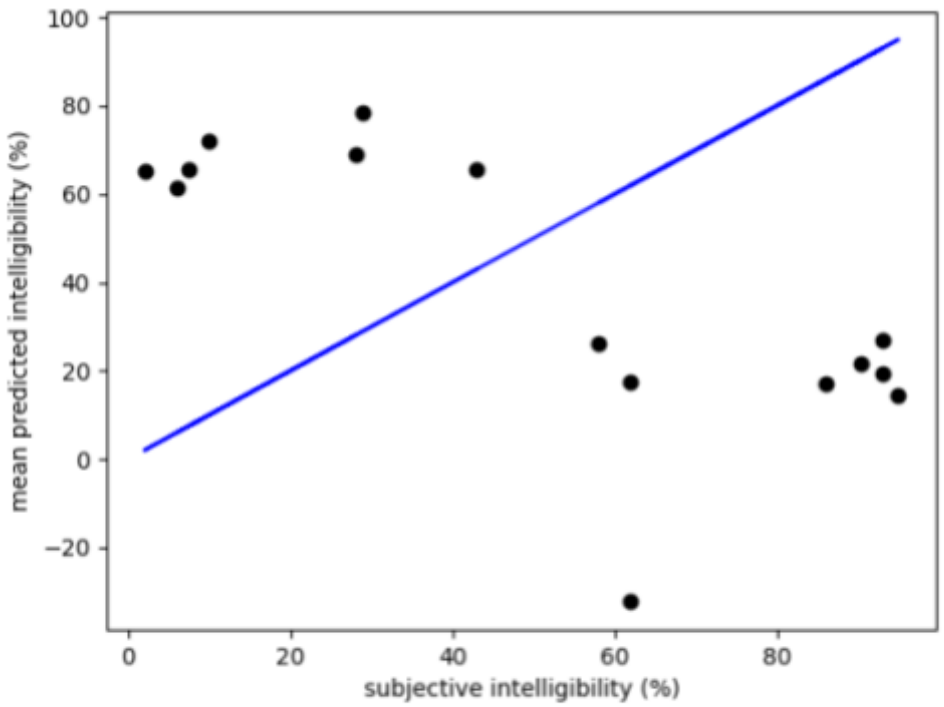

Figure 15. Estimated intelligibility versus subjective intelligibility obtained for the two-stage intelligibility assessment system when utterances are misclassified.

The performance of the system is severely affected by misclassified data. The change in feature weighting between the two severity groups is so stark as to completely remove or reverse any correlation once the labels are inverted. Performance is far worse than the single-stage system. The table below shows the results in more detail.

	R^2 Error	RMSE Error	Rs	Rp
low_mid	-19.48	470.71	0.00	0.39
mid_high	-12.31	270.60	0.71	0.35
Total	-2.21	377.32	-0.68	-0.74

Table 10. A more detailed look at the results of the Two-stage intelligibility classifier when utterances are misclassified.

By looking at the performance of the individual regression models, as well as the combination of the two, we can see that the performance of the *low_mid* model provided with *mid_high* utterances is far worse than the reverse case, though both are by far the worst performance from any test in this paper. The R^2 error in all cases is negative, indicating that the regression model performs worse than a horizontal line at the mean of the estimated scores. The correlation scores for the subsystem as a whole are also negative. This poor performance is concerning. Of course, the system was intentionally fed bad data at this point to provoke a bad result, but the increase in error is particularly large. There are some reasons not to worry, however.

As the performance is so bad, it is possible that an error could be detected. The system performance is worse than that of the single-stage implementation, an implementation that is not prone to this error. There is, thus, the potential for both systems to be used in parallel. If the single regression system produces a result that is incredibly different from the two-stage system, far greater than the error shown here, the classification label could be inverted and the objective intelligibility could be recalculated using the opposite model from the dual-regression subsystem.

Instead of attempting to solve a potential issue like this, there may be no issue at all. The classification system is 80% accurate and the system as a whole is designed to take several utterances and produce an intelligibility score which is the average of the scores calculated for each utterance. It would, of course, be naive to assume that the chances of several utterances from the same speaker being misclassified are independent of each other. By looking at the severity label misclassification rate for each speaker, the risk of a misclassified utterance greatly affecting the intelligibility score of a new speaker can be approximated.

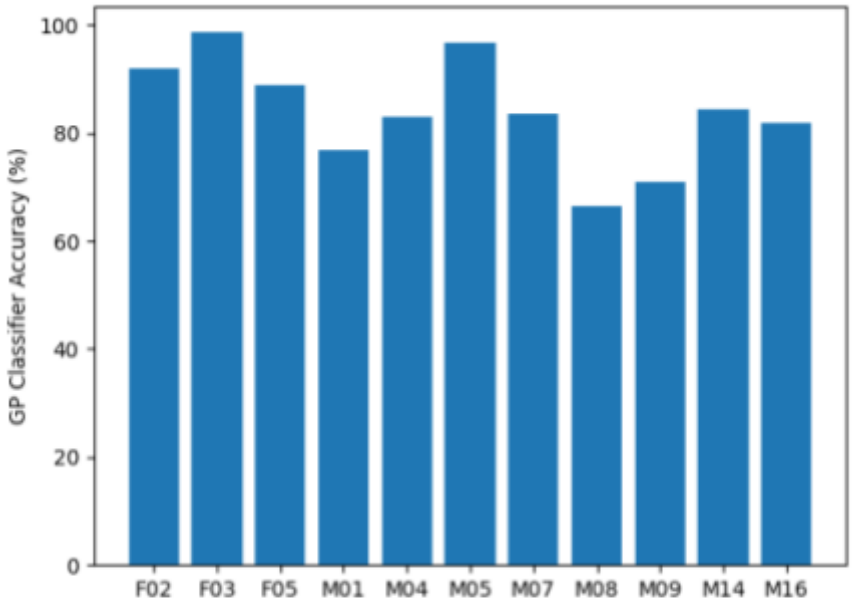

Figure 17. The accuracy results from each run of the classification subsystem when validated using the LOSO method

Figure 17 shows the classifier accuracy for the classifier subsystem using the Gaussian Process algorithm. The system was trained using the LOSO train-test split and the accuracy of the system for each iteration is shown. The consistency of the classifier's accuracy indicates that classification accuracy could be independent of the speaker and more dependent on the utterance or other factors. This is hard to say conclusively, as the dataset is too small to be confident, but it seems likely that by averaging the results for several utterances from one speaker, the impact of outliers will be reduced.

With this information, it is possible that misclassified utterances being sent to the wrong regression model might not be an issue. Outliers can be expected to occur roughly 20% of the time, independent of the speaker, and the values will be so different from the correctly classified utterances that identification and exclusion of outliers should be trivial.

8 . Summary and Conclusion

Over the course of this paper, a two-stage system for classifying spastic dysarthric speech by low_mid or mid_high severity and using one of two polynomial regression models to provide an objective estimate of the speaker's intelligibility has been created and tested. In this section, a final summary of the developed system will be given and compared against the original goals of the project laid out in Chapter 1 and Chapter 2. Comparisons will also be made to previously existing systems and the clinical potential for these systems will be discussed.

8.1 . Final System Specification

The final proposed system for providing objective estimates of the intelligibility of spastic dysarthric speech is a two-stage approach comprised of an initial severity classifier and dual polynomial regression models.

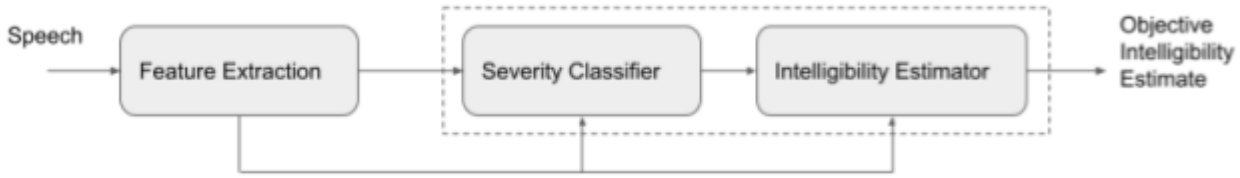

Figure 18. An overview of the system architecture for the novel objective intelligibility estimator

The following features are extracted from single-word utterances, forming the basis of objective estimates for each subsystem: fundamental frequency standard deviation, fundamental frequency range, voiced percentage, delta zeroth-order cepstral coefficient standard deviation, spectral tilt, and harmonic-to-noise ratio. The first four of these have been used successfully in previous systems, while the final two are unseen in directly comparable systems. Feature values are standardised by calculating the z-score, balancing the impact of each feature on distance-based algorithms.

	Precision	Sensitivity	F1-Score
mid_high	0.86	0.72	0.78
low_mid	0.79	0.90	0.84
accuracy			0.82

Table 11. The final performance results for the classifier subsystem

The classification subsystem uses a Gaussian Process binary classifier to broadly classify utterances as either *low_mid* intelligibility (0-40%) or *mid_high* (41-100%) intelligibility. The performance of the classifier is quite balanced, offering similar accuracy across both groups, but with a slight improvement inaccurately labelling *low_mid* intelligibility speakers. The overall accuracy of the classifier is more accurate than those found in previous literature using the same dataset by over 6%. Accuracy could be further improved by reducing the boundary between the labels to 25%, though a boundary this low negatively affects the performance of the polynomial regression models later.

Objective estimates of intelligibility are then calculated using two polynomial regression models, a 3rd-degree polynomial model trained on *low_mid* intelligibility utterances and a 2nd-degree polynomial model trained on *mid_high* intelligibility utterances. By training a regression model specifically for each group, the total overall RMSE error of the subsystem output is reduced by approximately 85% to just 6.19. This presents another improvement over previous works, which so far achieved an RMSE error of 6.33 at a minimum when using a combination of prosodic and cepstral features used here, as well as glottal parameters [34].

	R^2 Error	RMSE Error	Rs	Rp
low	78.24	6.85	1.00	0.96
high	86.04	5.07	0.6	0.97
all	97.00	6.11	0.96	0.99

Table 12. The final regression error results for both regression models and the overall subsystem

8.2 . Assessment Against Project Objectives

In Chapter 1, a set of objectives to be met in this project was established based on the background research and literature review undertaken. To assess the success of the project as a whole, the final system will be evaluated against each of the initial objectives.

The first objective was to create a system capable of objectively estimating the intelligibility of dysarthric persons from speech recordings. Using the UA-Speech database, the novel system presented in this work does indeed provide an estimate of speech intelligibility in persons with cerebral palsy and spastic dysarthria, to a degree of accuracy competitive with existing literature. Rather than the estimate being provided by a clinical speech pathologist or a panel of naive listeners whose subjective estimates are tested for congruence and averaged, the estimate provided here is the result of polynomial regression using features extracted directly from audio recordings.

The second objective was to create a system that leverages information presented by existing works as well as novel ideas based on research into spastic dysarthria directly. The system created here uses an architecture based on [34] to form the basis of the system. Through using a classifier and two regression models, a substantial improvement over a single-model system is achieved. The classification subsystem was created through research into the use of multiple-classifier frameworks over traditional methods [36]. Previous work into multiple-classifier systems focuses on the use of different classifiers across several speech tasks, however. This introduces a novelty to the system created in this work which applies the framework to a single speech task. Furthermore, new features were researched to great effect, with correlation measurements for these novel features exceeding those recommended by previous works.

Finally, the third aim was to assess the clinical need and potential for the developed system. Chapter 1 covered the clinical need for such a system, denoting that current subjective assessments of speech intelligibility are potentially prone to bias and require the physical presence of the patient and a specialist in the same room. Not only is this time consuming but it would be hard for this paper to ignore the context of the Covid-19 pandemic. In 2020 and 2021, contact between unrelated persons was limited to avoid the risk of contracting the virus, which presents an increased risk to sufferers of cerebral palsy [69]. By utilising technology like this, using 5 or more words spoken into a smartphone, a faster and more efficient method of clinical assessment is possible [7].

9 . Project Planning and Management

This chapter will begin by discussing and reflecting upon the scheduling of the project, considering both fixed and self-imposed deadlines. As with most major projects, especially those devised by those with little experience managing projects of this scale, it was not an entirely smooth experience. How the aims of the project changed over its duration will be detailed and explained in this chapter.

9.1 . Initial Research and Timeline

In the initial report submitted for this project, a timeline was proposed for the project and the results of initial research and literature reviews were presented. This timeline was developed as a rough guide before any work had taken place and is presented below in Figure 19.

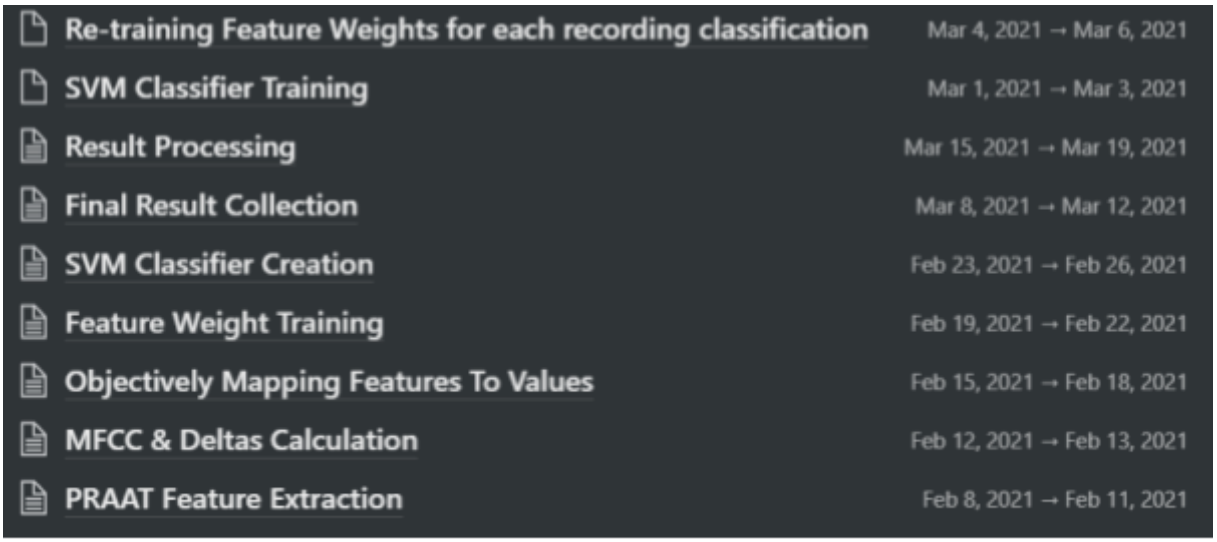

Figure 19. The initial timeline for the project.

The initial timeline was based on the assumption that the Parkinson's dataset would be used, only one classifier would be implemented, and does not include any pilot experiments for future work. Furthermore, the time assigned to each step of the project was decided having never implemented a machine learning system before. As such, the actual project timeline was far from this initial outline.

9.2 . Timeline Revision for the UA-Speech Dataset

When the support vector machine classifier and polynomial regression model were initially developed and initial testing was undertaken, their poor performance highlighted the fact that the Parkinson's Dataset may not be fit for use in this project. At the beginning of March, the UA-Speech dataset was procured and the project was re-tooled to fit the new data. This immediately led to better and more consistent results. The confidence this provided inspired the scope of the project to be expanded to include multiple classifiers and several pilot experiments for future work. The revised timeline, with dates adjusted to reflect a more accurate duration for each task, is shown in the table below.

Name	Date
Other Dysarthria Testing	May 3, 2021 → May 7, 2021
Final Result Collection	Apr 12, 2021 → Apr 16, 2021
Report Writing	Mar 29, 2021 → May 30, 2021
Initial Result Collection	Mar 26, 2021 → Mar 29, 2021
Whole System Retraining	Mar 19, 2021 → Mar 25, 2021
Other Classifier Implementation	Mar 14, 2021 → Mar 18, 2021
SVM and Regression Re-training and results Analysis	Mar 10, 2021 → Mar 12, 2021
UA-Speech Dataset Procurement and Management	Mar 5, 2021 → Mar 9, 2021
Parkinsonian Dataset Results Analysis	Mar 2, 2021 → Mar 4, 2021
Regression Model Training	Feb 26, 2021 → Mar 1, 2021
Regression Model Development	Feb 22, 2021 → Feb 25, 2021
SVM Classifier Training	Feb 18, 2021 → Feb 20, 2021
SVM Classifier Creation	Feb 15, 2021 → Feb 17, 2021
Parselmouth Feature Implementation	Feb 10, 2021 → Feb 13, 2021
Filesystem Management and Data Structure Development	Feb 8, 2021 → Feb 12, 2021

Figure 20. The revised final project timeline

10 . Future Work

The project undertaken and described in this paper leaves plenty of room for improvement and further work in the future. This chapter will briefly cover some of the areas not fully explored in this project, as well as potential extensions to the final system.

10.1 . Glottal features

In a paper heavily referenced by this work, researchers used the same dataset and a similar system design to create objective intelligibility estimates. Performance of the system was exceeded by this work, though some of the features implemented in the paper which were not tested here are stated to perform exceptionally well [34]. The paper demonstrates the use of two sets of glottal parameters: time-domain and frequency-domain parameters (referred to as Glottal-1) and parameters based on principal component analysis (referred to as Glottal-2). Systems developed using these features are compared to systems utilising prosodic and cepstral features implemented here and are shown to be superior in their performance.

	Classification Accuracy (Test data)		Classification Accuracy (Validation data)	
Feature set	Without feature selection (%)	With feature selection (%)	Without feature selection (%)	With feature selection (%)
Falk features	62.69	N/A	62.93	N/A
Glottal-1	71.15	74.08	71.74	75.29
Glottal-2	66.15	67.06	67.70	69.51
Glottal-1 + Glottal-2	68.51	74.90	68.85	75.75

Figure 21. A table of results comparing classifier accuracy when using glottal features vs typical prosodic features

By implementing glottal features alongside harmonic-to-noise ratio and spectral tilt, features not utilised by [34], it is possible that a further improvement could be achieved, bolstering the already impressive performance of the novel system developed in this work.

10.2 . A Model for Non-Spastic Dysarthrias

An area where this work is lacking so far is that the model is specific to only one type of dysarthria: spastic type dysarthria. The original intention of this project was to be applied to Parkinson's disease, but the dataset made available was not suitable, as previously discussed. The system developed here is not expected to work well with other dysarthrias, given the reduction of effectiveness even when the regression models make predictions on different severities of the same dysarthria type. However, extending the model in two ways could make for a much more clinically useful product by integrating dysarthria type detection and retraining regression models to accommodate for other dysarthria types.

10.2.1 . Evaluating the Current System with Non-Spastic Type Dysarthrias

In the US-Speech dataset, speech recordings are provided for 4 subjects with non-spastic type dysarthria. Their diagnoses and biographical data are provided in the table below.

Speaker	Age	Speech Intelligibility (%)	Dysarthria Diagnosis
M10	21	93	Mixed
M11	48	62	Athetoid
M12	19	7.4	Mixed
F04	18	62	Athetoid

Table 13. Biographical data on the non-spastic dysarthric speakers in the UA-Speech database [26]

To evaluate the performance of the current system against these non-spastic type dysarthrias, recordings from these speakers will split into two sets using the common-uncommon split method used previously. Although there is no need to create a training and validation set, it is possible for the system to perform better using the common word set as it has previously seen all the words in this set. The words were spoken by different speakers with different diagnoses but it's important to separate the sets to recognise any potential for bias.

Each set of recordings will be fed through the final revision of the system, including the same feature mean and standard deviation values used for standardising extracted features. This is a key issue in using z-scores for machine learning tasks. If the training data is not fully representative of the range of values each feature can be expected to exhibit, the means and standard deviations calculated will be different from the values found in the real world. This can seriously skew the results if a new data point is observed outside of the expected range, giving it a much more extreme z-score which could cause that feature to dominate the performance of the system. This is exactly what z-scores are used to avoid, and also the precise reason why the system presented in this work, as it stands currently, is not expected to be accurate in the classification and regression of speakers with non-spastic dysarthria.

Speaker	Common Word Accuracy	Uncommon Word Accuracy
F04	0.28	0.34
M10	0.87	0.96
M11	0.34	0.47
M12	0.93	0.95

Table 13. Classification accuracy results for the common and uncommon word sets for non-spastic dysarthric speakers

It should be noted that for speakers with a mixed-type dysarthria diagnosis, no further information is provided as to the details of the diagnosis. If the speakers exhibit many symptoms of spastic-type dysarthria, they may receive more accurate values than speakers with strictly athetoid-type. As the table above shows, this may be the case. Accuracy scores for utterances spoken from M10 and M12, the two speakers with ‘mixed’ diagnoses, are incredibly high, some of the highest seen for any participants in the study. This indicates that these speakers may present many of the speech symptoms associated with spastic dysarthria, as well as other symptoms consistent with other dysarthria types.

Accuracy for the athetoid dysarthric speakers, on the other hand, is very low, as was expected. Though some of the features used, such as voiced frame percentage, apply to both spastic and athetoid dysarthria, they still present to different degrees in speakers of differing dysarthria. Because of this, features that might have high correlations with both dysarthria types can still cause inaccurate classifications, as the influence of that feature over the new dysarthria type, the type the system isn't trained for, is improperly weighted against the other features.

Speaker	Subjective Intelligibility (%)	Training Data Mean Predicted Intelligibility (%)	Training Data RMSE Error	Test Data Mean Predicted Intelligibility (%)	Test Data RMSE Error
F04	62	87.52	26.24	85.98	26.34
M10	93	87.94	6.79	84.71	10.76
M11	62	85.42	24.33	83.29	22.47
M12	7.4	13.92	9.42	14.75	10.51

Table 14. Regression accuracy results for the common and uncommon word sets for non-spastic dysarthric speakers

Further to this, we can see in the table above that the regression models show similar results. For the speakers with athetoid-type dysarthria, the RMSE error of the regression models across both test and training data is far higher than seen in the mixed-type speakers. The RMSE Error observed using the training utterances for speaker M10 is comparable to that of the system during validation with spastic-type dysarthric speakers, but aside from this, results from the mixed-type speakers in general still show somewhat poorer performance than previously seen. This contrasts the classification results, which for both mixed-type speakers are on the higher end of the expected accuracy of the system.

This could be partly due to the regression models being far more susceptible to subtle differences in the weighting of feature values. As the previous subsystem utilises binary classification, the threshold for error is higher than a regression model which calculates a specific value, using each feature in the process. Not only does this allow for easier quantification of the error but it makes the system more comparable to distance-based classifiers in which large differences in individual feature values can dominate classification and cause far more errors than in other algorithms.

The results of this small pilot experiment show the potential for an extension to the main system demonstrated in this paper. There is clear potential for a grander system to be trained if more data on athetoid-type dysarthric speakers, or speakers with Parkinsonian and other dysarthrias, were available. The system could potentially be trained to identify the speaker's dysarthria diagnosis, classify the severity and then provide an accurate measure of speech intelligibility, which a healthcare provider can then use for easy and remote long-term monitoring of the condition.

10.2.2 . Evaluating the Current System with Parkinsonian Dysarthria

The original aim of the system was for use with the Parkinson's Dataset. This dataset had the issue of being too homogeneous, not providing enough data for speakers with *low_mid* intelligibility, and so the UA-Speech dataset was used in its place. The results of the previous experiment applying the system to non-spastic dysarthria shows that even with the same speech task, with utterances recorded under the same conditions, the system needs re-training to calculate new weights to balance the significance of features to accurately detect and quantify other types of dysarthria. This alone does not bode well for the Parkinson's Dataset, which also uses a different speech task.

The recordings from the Parkinson's Dataset use prolonged speech extracts, reading around 30-50 seconds of speech, as opposed to the single words uttered in the UA-Speech dataset. To make the recordings more compatible, the Parkinson's recordings were scoured for words utilised in the UA-Speech dataset. These words were then cut out as cleanly as they could be and compiled into a small dataset. Utterances from two speakers were extracted, one with mid_high intelligibility and one with low_mid intelligibility, as noted in the biographical data for each speaker provided with the Parkinson's Dataset. The results of this experiment are noted in the tables below.

Speaker	Common Word Accuracy	Uncommon Word Accuracy
RA_1YR_ON	0.23	0.18
AP_PRE_OFF	0.30	0.21

Table 15. Classification accuracy results for the common and uncommon words extracted from Parkinsonian dysarthric speakers reading a short text extract

Speaker	Subjective Intelligibility (%)	Training Data Mean Predicted Intelligibility (%)	Training Data RMSE Error	Test Data Mean Predicted Intelligibility (%)	Test Data RMSE Error
RA_1YR_ON	13	-14.43	43.87	-11.70	36.78
AP_PRE_OFF	98	79.97	25.35	71.54	28.76

Table 16. Regression accuracy results for the common and uncommon words extracted from Parkinsonian dysarthric speakers reading a short text extract

The performance of the system on these recordings is very poor, as expected. In a study previously outlined in this paper, it has been shown that classifier accuracy is affected by speech task, with different classifiers performing better or worse on differing speech tasks even when the same features are extracted [36]. This is exacerbated by the fact that the classifier here has been pre-trained on a different speech task and a different dysarthria type. Performance here is roughly comparable to the system's performance on athetoid-type dysarthria, if slightly worse. It should be noted however that a significantly smaller dataset was used for this experiment, as matching words from both studies had to be manually searched for and cut out. Even so, performance is clearly not satisfactory and a new system would have to be trained, if the original framework developed here could be reused at all effectively. It's possible that new features would need to be extracted and a new classifier would be found to be optimal with parkinsonian dysarthric speech.

10.3 . Optimisation of the algorithm

As well as extending the functionality of the current system, further optimisation of the algorithm could allow for expert human-like diagnosing and clinical evaluation. As it stands, the system presented here extends ideas presented before by refining the classification method and evaluating the potential for new features. By combining new and old features, expanding training data and utilising more advanced optimisation techniques, the strength of the system could be increased further. Some of the algorithm optimisation techniques available but not used in this paper are outlined below.

10.3.1 . Algorithmic Hyperparameter Tuning

For each machine learning algorithm used here, there are several hyperparameters that affect the performance of the system. For example, how many trees are used in a random forest, what should be the value for C and sigma in a support vector machine, what degree polynomial should be used in polynomial regression, or what should k be for nearest neighbours. In this paper, these values were decided through manual search. Scikit Learn provides default values for these parameters when instances of the classifiers are initialized and these values were used initially. Each variable was then varied over a small range and the most optimal values were chosen. This is a very basic approach relying heavily on intuition and the default values being already close to optimal.

A better approach is known as grid search. In grid search, a range of values is defined for each hyperparameter and each combination of values is tested. For 1 model with 4 hyperparameters and 20 values per hyperparameter, this results in 160,000 trials. This is then multiplied by 14 to account for each of the 11 classifiers used and the 3 regression models. This is obviously extremely computationally intensive and with the average trial taking 7 minutes, training could take years. To explore this grid in a reasonable time, sophisticated algorithms are employed to search and traverse only the most relevant parts of this grid. Many of these algorithms, much like many machine learning models, are biologically inspired.

With such large feature spaces, it's probable that much of that space is unimportant. There will be either a single point, or an array of points in systems where multiple objectives are to be considered, which is most optimal and there are paths leading either to or away from these points. Algorithms like the moth-flame algorithm have been developed to mimic search methods developed through millions of years of evolution and apply the traversal strategy to grid search [70]. By leveraging evolutionary algorithms, named as they have been naturally optimised through the evolution of the species on which the algorithm is based, the optimal hyperparameter front in a grid search can be found in far less time. Applying this methodology to the system presented in this paper could yield hyperparameter values which are far more optimal, ultimately creating a more accurate system.

10.3.2 . Receiver Operating Curves

Receiver Operating Curves are plots of specificity against sensitivity for machine learning classifiers. Viewing the data in this way allows for data scientists to strike an appropriate balance between the two values, depending on the exact use case for their system. In this paper, this process was not undertaken as the classifier does not directly present information to the user, the result is used purely internally, and when optimising for accuracy using the default values presented by Scikit Learn, an appropriate balance had already been struck. In future work, when the effect of misclassifications is more fully understood or either specificity or sensitivity become more valuable, receiver operating curves would allow for better visualisation of the consequences of each of these values when developing the system. An example of this is shown in the figure below [71].

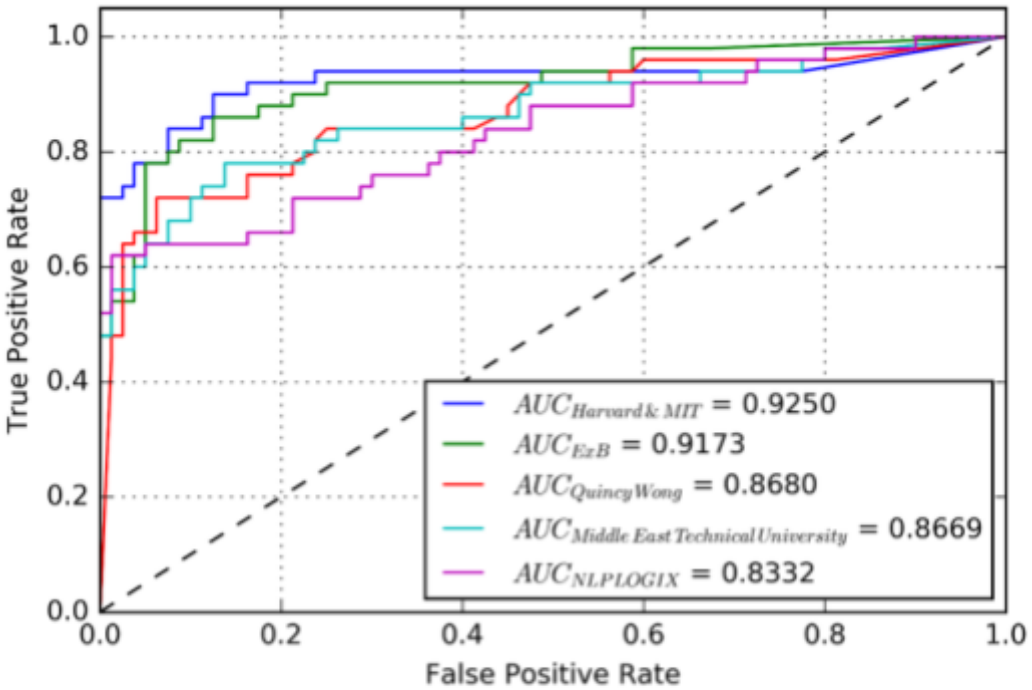

Figure 22. A receiver operating curve comparing the performance of several machine learning classification systems for classifying metastatic breast cancer

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